SEC P vs NP log 2026-05-17

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-18 17:00 UTC

SEC P vs NP — daily log 2026-05-17

43 cycles recorded

GCT Lie-Stabilizer Codimension Linearly Bounds ACC^0 Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Geometric Complexity Theory: Lie-algebra stabilizers of multilinear polynomial extensions in the Mulmuley-Sohoni-Bürgisser-Ikenmeyer framework × $ACC^0[m]$ circuit size for Boolean functions (Sipser-like targets)
- **Recorded**: 2026-05-17 00:01 UTC
- **Entry ID**: `83d26c2ef069`

Statement

For a Boolean function f:{-1,+1}^n -> {-1,+1}, let F_f(x) = Σ_{S⊆[n]} f̂(S) · Π_{i∈S} x_i be its multilinear Walsh extension over Q, let L(F_f) := {A ∈ gl_n(Q) : Σ_{i,j} A_{ij} x_j ∂_i F_f ≡ 0 in Q[x_1,…,x_n]} be its GCT Lie stabilizer, and set γ(f) := n^2 − dim_Q L(F_f). Conjecture: there is an absolute constant C ≤ 10 such that every $ACC^0[m]$ circuit of size s computing f satisfies γ(f) ≥ n^2 − C·s. A single $ACC^0[m]$ circuit of size s computing some f with γ(f) < n^2 − 10·s falsifies the conjecture.

Rationale

GCT's symmetry-characterization principle says polynomials whose stabilizer Lie algebras exceed generic dimension are atypical; we invert this to claim small ACC^0 circuits cannot accidentally synthesize multilinear extensions of anomalously high symmetry. The Lie stabilizer is GL_n-equivariant rather than truth-table-dense, so it dodges Razborov–Rudich's 'large + constructive' clause, and it is computed from the GL_n action on polynomials, not from black-box algebraic extensions — placing the invariant outside the algebrization and relativization regimes that GCT was designed to escape.

Novelty

- Judge: `NOVEL` over 11 arXiv hits

Top hits consulted: - [1212.5380v2] On properties of principal elements of Frobenius Lie algebras - [1606.05050v1] Proof Complexity Lower Bounds from Algebraic Circuit Complexity - [2411.11095v3] Invariant theory and coefficient algebras of Lie algebras - [1904.05483v2] Parallels Between Phase Transitions and Circuit Complexity? - [1504.04131v2] Formulas for the Walsh coefficients of smooth functions and their application to bounds on the Walsh coefficients

Empirical Test

- exit code: `1`, elapsed: `0.11s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_698adb63.py", line 102, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_698adb63.py", line 86, in run_trial
    gamma = gaussian_elimination(system)
            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_698adb63.py", line 65, in gaussian_elim
    if matrix[i][i] == 0:
       ~~~~~~~~~^^^
IndexError: list index out of range
```

Judge reasoning

The test crashed with an IndexError in gaussian_elimination before any of the 480 trials or reference-function checks produced slack data, so the pre-registered support condition cannot be evaluated. Per Rule 2, a crash yields INCONCLUSIVE. | next: Fix the Gaussian elimination routine (guard against out-of-range pivot indexing and handle rectangular/short matrices) and rerun the full (n,s)-grid plus reference-function suite to obtain actual slack statistics.

2-adic Newton Slopes of DISJ Sub-blocks Scale as √n

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: p-adic analysis (Newton polygons over Q_2 of integer characteristic polynomials) × Randomized communication complexity of DISJOINTNESS via average-case tensor invariants on sub-blocks
- **Recorded**: 2026-05-17 00:26 UTC
- **Entry ID**: `efd166108501`

Statement

Let M_n ∈ {-1,+1}$^{\{2}$n × 2^n} be the signed disjointness matrix M_n[x,y] = (-1)^{[x∩y≠∅]} where x,y are interpreted as subsets of [n]. For each n, draw uniformly random row set R and column set C with |R|=|C|=k=min(2n,40), form A := M_n[R,C], compute its exact integer characteristic polynomial p(t)=Σ_{i=0}^{k} c_i t^i, and define the 2-adic Newton slope count ν_2(A) := number of distinct slopes in the lower convex hull of {(i, v_2(c_i)) : c_i ≠ 0} where v_2 is the 2-adic valuation (v_2(0):=+∞). Conjecture: E_{R,C}[ν_2(M_n^{R,C})] = Θ(√n); moreover for every n ≥ 8 the DISJ median exceeds the median of ν_2 for an i.i.d. ±1 matrix of the same shape by a factor ≥ 1.5, and a single (n, seed) pair where this median ratio inverts falsifies the conjecture.

Rationale

The subset semilattice underlying DISJ injects 2-adic congruence structure into every sub-block (parity of intersection size acts like a Z_2-valued character), so subdeterminants accumulate distinguished 2-adic valuations that random Bernoulli matrices lack — a property that should propagate, via Yao's principle, into a lower bound on randomized communication. Newton polygons over Q_2 are not preserved by polynomial extension to F_2[t]/⟨t^2-t⟩, evading the algebrization barrier, and the invariant is structural rather than 'large+useful on truth tables', evading Razborov-Rudich. The bridge between p-adic Newton-polygon geometry and communication complexity has essentially no prior literature, while the average-to-worst-case framing (E[ν_2] ↦ worst-case CC lower bound) follows the Impagliazzo-Wigderson template.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `34.76s`

```
tances_tested': 5, 'conjecture_holds': False, 'counterexample': 'Ratio 1.0 < 1.5 for seed 547'}
TRIAL: {'metric_name': 'median_newton_slope_count', 'metric_value': 2, 'instances_tested': 5, 'conje
TRIAL: {'metric_name': 'median_newton_slope_count', 'metric_value': 2, 'instances_tested': 5, 'conje
TRIAL: {'metric_name': 'median_newton_slope_count', 'metric_value': 2, 'instances_tested': 5, 'conje
TRIAL: {'metric_name': 'median_newton_slope_count', 'metric_value': 2, 'instances_tested': 5, 'conje
TRIAL: {'metric_name': 'median_newton_slope_count', 'metric_value': 2, 'instances_tested': 5, 'conje
TRIAL: {'metric_name': 'median_newton_slope_count', 'metric_value': 2, 'instances_tested': 5, 'conje
TRIAL: {'metric_name': 'median_newton_slope_count', 'metric_value': 2, 'instances_tested': 5, 'conje
TRIAL: {'metric_name': 'median_newton_slope_count', 'metric_value': 2, 'instances_tested': 5, 'conje

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b68324ac.py", line 136, in <module>
    failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                   ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b68324ac.py", line 136, in <genexpr>
    failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                        ~^^^^^^^^
KeyError: 'seed'
```

Judge reasoning

Safety rail: critic_challenge_falsified | original: Multiple trials report conjecture_holds=False with ratio 1.0 < 1.5 (e.g., seeds 547, 593, 631, …), satisfying the pre-registered falsification condi | next: Re-run with corrected seed-logging, stratify by $n \in \{10,14,18,22\}$, and check whether the ν_2 statistic saturates at small k=min(2n,40) — perhaps a normalized slope-density (ν_2/k) would better capture any DISJ-vs-random gap.

Cyclic Burnside Orbit Count Lower-Bounds AC^0 Size for PARITY

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Burnside-ring / equivariant orbit counting of cyclic group actions on labeled circuit DAGs (a corner of combinatorial representation theory rarely applied to circuit complexity) × AC^0 circuit size for PARITY_n, with the lifting-style intuition that gadget symmetry constrains the lifted communication matrix
- **Recorded**: 2026-05-17 00:49 UTC
- **Entry ID**: `80d12cf7a27e`

Statement

Let C be an AC^0 circuit over inputs x_0,…,x_{n-1} computing PARITY_n with n a prime ≥ 3, depth(C)=d and size(C)=s, and let σ be the cyclic shift σ(i)=i+1 mod n acting on each gate g of C by simultaneously relabeling every input wire x_i appearing in g's recursive expansion to x_{σ(i)}. Define ψ(C) as the number of ⟨σ⟩-orbits of gates of C under the equivalence g ∼ h iff some σ^k maps the lex-min canonical (op-type, sorted-children) form of g to that of h. Then for every such C computing PARITY_n we conjecture ψ(C) ≥ (1/(4d)) · log_2(s), refuted by any single PARITY-computing AC^0 circuit whose ratio 4·d·ψ(C)/log_2(s) drops below 1.

Rationale

Lifting theorems translate query lower bounds into communication lower bounds by composing the function with a symmetry-breaking gadget; dually, when the target function (PARITY) is itself shift-invariant, a small circuit must reuse few genuinely distinct sub-computations, so the Z_n-orbit count of its gates becomes a Burnside-style proxy for the work that random restrictions are usually invoked to expose. Because ψ is defined on the circuit DAG rather than on the truth table, the invariant sits outside the Razborov–Rudich natural-proofs barrier; because the construction is purely combinatorial (no polynomial extension of the Boolean ring) it does not algebrize, matching the LIFTING approach's safe-direction profile.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [2304.02770v2] Tight Correlation Bounds for Circuits Between AC0 and TC0 - [2504.19966v3] Quantum circuit lower bounds in the magic hierarchy - [1708.00951v2] Canonical Heights and Monomial Maps: On Effective Lower Bounds for Points with Dense Orbit - [1904.05483v2] Parallels Between Phase Transitions and Circuit Complexity? - [2005.06373v2] Counting Schur Rings over Cyclic Groups of Semi-prime Order

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c17bcfb6.py", line 225, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c17bcfb6.py", line 180, in run_trial
    psi = compute_psi(circuit, n)
          ^^^^^^^^^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c17bcfb6.py", line 139, in compute_psi
    gate_hashes[gate] = hash_gate(gate)
                        ^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c17bcfb6.py", line 126, in hash_gate
    return hash(gate)
           ^^^^^^^^^^

TypeError: unhashable type: 'list'
```

Judge reasoning

The test crashed with a TypeError (unhashable list) in hash_gate before any trial completed, so no r values were produced and the pre-registered support condition could not be evaluated. | next: Fix the canonicalization in hash_gate by converting children lists to tuples (or using a frozen canonical form) before hashing, then re-run the full 2160-circuit grid.

Cauchy Mean Width of Minterm Hull Lower-Bounds Monotone k-CLIQUE

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Convex geometry (Cauchy mean-width / intrinsic 1-volume of polytopes in R^N, in the spirit of Schneider's Brunn–Minkowski theory, rarely connected to circuit complexity) × Karchmer–Wigderson / monotone DNF size and formula complexity for the k-CLIQUE indicator
- **Recorded**: 2026-05-17 01:28 UTC
- **Entry ID**: `65d1ffdf3cc6`

Statement

For a monotone Boolean function f on N variables, let M(f) ⊂ {0,1}^N be its set of minterms (minimal satisfying assignments viewed as 0/1 vectors) and let K(f) := conv(M(f) ∪ {0}) ⊆ [0,1]^N. Define μ(f) := MW(K(f))^2, where MW(K) = 2·E_{u~Unif(S^{N−1})}[max_{x∈K}⟨u,x⟩] is the Cauchy mean width. We conjecture: (i) for every monotone DNF representation of f with s terms, μ(f) ≤ C_1·log(s+1)·(log N+1); (ii) for the k-CLIQUE indicator on K_v with k=⌈$\log_2$ v⌉ (so N = v(v−1)/2), μ(f) ≥ C_2·v, for absolute constants C_1,C_2>0. A single instance with μ(f) > C_1·log(s+1)(log N+1), or a k-CLIQUE indicator with μ < C_2·v at v ≤ 9, falsifies the conjecture.

Rationale

Monotone DNFs are unions of axis-aligned subcubes, so conv(M(f)) is the convex hull of at most s lattice vectors of bounded coordinate sum, and its mean width should grow only logarithmically with the number of generators (a Sudakov/Talagrand-style upper bound for sparse 0/1 hulls). The k-CLIQUE minterm cloud, in contrast, lives on the (k choose 2)-th slice and is rotationally rich because v-vertex automorphisms act transitively on edges, forcing the supporting hyperplanes to spread evenly over many directions and inflating mean width linearly in v. If both bounds hold, any monotone DNF for k-CLIQUE has s ≥ 2^{Ω(v/log v)}, a Razborov-flavor separation via a purely metric-geometric witness that bypasses approximator combinatorics.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `33.94s`


```
me": "mean_width_squared", "metric_value": 1.136003662687513, "instances_tested": 17, "conjecture_ho
CLIQUE with v=4 has mu=1.136003662687513 < 0.5*v=2.0", "seed": 421}
TRIAL: {"metric_name": "mean_width_squared", "metric_value": 1.1857866298955364, "instances_tested"
CLIQUE with v=4 has mu=1.1857866298955364 < 0.5*v=2.0", "seed": 463}
TRIAL: {"metric_name": "mean_width_squared", "metric_value": 1.1927318541604457, "instances_tested"
CLIQUE with v=4 has mu=1.1927318541604457 < 0.5*v=2.0", "seed": 503}
TRIAL: {"metric_name": "mean_width_squared", "metric_value": 1.1396899464107255, "instances_tested"
CLIQUE with v=4 has mu=1.1396899464107255 < 0.5*v=2.0", "seed": 547}
TRIAL: {"metric_name": "mean_width_squared", "metric_value": 1.1832069622303316, "instances_tested"
CLIQUE with v=4 has mu=1.1832069622303316 < 0.5*v=2.0", "seed": 593}
TRIAL: {"metric_name": "mean_width_squared", "metric_value": 1.1742179582193337, "instances_tested"
CLIQUE with v=4 has mu=1.1742179582193337 < 0.5*v=2.0", "seed": 631}
TRIAL: {"metric_name": "mean_width_squared", "metric_value": 1.1635078759135067, "instances_tested"
CLIQUE with v=4 has mu=1.1635078759135067 < 0.5*v=2.0", "seed": 677}
TRIAL: {"metric_name": "mean_width_squared", "metric_value": 1.1855075477194135, "instances_tested"
CLIQUE with v=4 has mu=1.1855075477194135 < 0.5*v=2.0", "seed": 727}
TRIAL: {"metric_name": "mean_width_squared", "metric_value": 1.172332125064841, "instances_tested":
CLIQUE with v=4 has mu=1.172332125064841 < 0.5*v=2.0", "seed": 773}
TRIAL: {"metric_name": "
```


Judge reasoning

Safety rail: critic_challenge_falsified | original: Every k-CLIQUE trial at v=4 yields μ ≈ 1.17, well below the falsification threshold 0.25·v = 1.0… actually 0.25·4=1.0 so this passes the 0.25·v floo | next: Rescale the lower bound to μ(f) ≥ C_2·v/log v or test the conjecture with normalized mean width MW(K(f))/√N to remove metric-saturation artifacts at small v.

Costas Displacement Coincidence Lower-Bounds AC^0 Size for PARITY

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Costas array / Welch–Costas displacement combinatorics (Golomb–Costas radar arrays and B_2/Sidon-deficiency theory of integer sets) — a corner of additive combinatorics developed for radar/sonar that has essentially no presence in circuit complexity literature × AC^0 depth-d circuit size for PARITY_n (Hastad-style lower bounds, viewed through the lens of bottom-gate support designs)
- **Recorded**: 2026-05-17 02:01 UTC
- **Entry ID**: `6bb89da9ec2e`

Statement

Let C be a depth-d AC^0 circuit on n inputs with bottom-level (depth-1) AND/OR gates having literal-supports S_1,…,S_m ⊆ [n]. Define the Costas displacement defect κ(C) = log_2(1 + max_{δ ∈ {1,…,n-1}} Σ_{i=1}^{m} |{(a,b) ∈ S_i × S_i : a > b and a−b ≡ δ (mod n)}|). We conjecture that for every C that computes PARITY_n exactly, κ(C) ≥ (1/4) · n^{1/(d−1)}, and consequently size(C) ≥ 2^{Ω(κ(C))}. A single depth-d AC^0 circuit that computes PARITY_n while having κ(C) < (1/4) · n^{1/(d−1)} refutes the conjecture.

Rationale

PARITY is translation-invariant on the Boolean cube, so any AC^0 circuit computing it must spread its bottom-level supports densely across all cyclic displacements — exactly the property that Costas/Welch arrays minimize. Replacing Hastad's random restriction by a Costas-style design-theoretic lower bound turns AC^0 PARITY bounds into a question about Sidon-deficiency of support multisets. The invariant depends only on the circuit's bottom-support combinatorics (not the function's truth table), so it sidesteps Natural Proofs, and uses no ring operations, sidestepping algebrization.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.18s`

```
TIMEOUT after 240s
```

Judge reasoning

The test timed out after 240s (returncode 124) without producing a RESULT line, so neither the support nor falsification condition could be evaluated. | next: Reduce the parameter sweep (e.g., cap n at 20 and d=2) or parallelize per-seed evaluation to produce data within the timeout.

Vertex Star-Discrepancy Equals Spectral Norm for XOR Comm

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Geometric discrepancy theory (vertex-restricted L∞ star discrepancy of {0,1}^n point sets, in the Chazelle–Matousek tradition of axis-parallel box discrepancy) × Randomized two-party communication complexity of XOR-functions f(x,y)=g(x⊕y) accessed through the discrepancy-method lower bound disc(M_f)=||ĝ||_∞
- **Recorded**: 2026-05-17 02:29 UTC
- **Entry ID**: `d27e7e45b29b`

Statement

For any non-constant g:{0,1}^n→{−1,+1}, let P_g={z∈{0,1}^n : g(z)=−1} and define the lattice star discrepancy D**lat(g) := max*{a∈{0,1}^n} | |P_g ∩ [0,a]⊓|*/2^n − (|P_g|/2*$^{n)\cdot(2}$*{popcount(a)}/2^n)* *|, where [0,a]*⊓={z∈{0,1}^n : z≤a coordinatewise}; let ||ĝ||∞ *:= max*{S⊆[n]} |ĝ(S)| be the spectral norm. Conjecture: there exist absolute constants c_1=1/8 and c_2=8 such that for every n≥4 and every non-constant g, c_1·||ĝ||_∞ ≤ D**lat(g) ≤ c_2·*||*ĝ*||∞·log_2(n+1). A single (n,g) pair violating either inequality refutes the conjecture.

Rationale

The discrepancy lower bound on randomized XOR-communication is exactly log(1/||ĝ||∞*), yet* ||*ĝ*||∞ is a global Fourier quantity with no geometric face. Vertex star discrepancy is the canonical Chazelle–Matousek irregularity measure of a point set inside [0,1]^n; restricting test boxes to lattice corners makes it computable in O(n·2^n) via the zeta transform. If D**lat* ≈ ||*ĝ*||∞, this gives the first concrete geometric (axis-parallel box) certificate of XOR-communication hardness, opening a Banaszczyk/Kashin-style geometric attack on unsaturated R(DISJ)-like bounds.

Novelty

- Judge: `NOVEL` over 2 arXiv hits

Top hits consulted: - [s2:1704.02537] Dual polynomials and communication complexity of XOR functions - [s2:19311e592db3bd54ad8685ae0d27b815c87be] Dual polynomials and communication complexity of XOR functions

Empirical Test

- exit code: `124`, elapsed: `240.03s`

```
TIMEOUT after 240s
```

Judge reasoning

The test timed out after 240s (returncode=124) without producing a RESULT line or any trial data, so neither the support nor falsification condition can be evaluated. | next: Reduce the maximum n or restrict the enumeration over a∈{0,1}^n (e.g., sample or use a smarter combinatorial bound) so the 210 trials complete within the time budget.

Operator-SoS 4-Trace Gap Lower-Bounds Blocks-Order Read-Twice BP for IP_2

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Sum-of-Squares / Lasserre hierarchy on operator-valued layer-product tensor moments (Putinar–Helton-style non-commutative SoS pseudo-trace invariants applied to layer-difference matrices of branching programs — a corner combining moment-SoS literature with BP structural invariants, with <5 papers on the BP side) × Read-twice oblivious branching program size for IP_2 (the open BP_READTWICE separation of Borodin–Razborov–Smolensky 1993)
- **Recorded**: 2026-05-17 02:59 UTC
- **Entry ID**: `0ecd5f06ea81`

Statement

Let P be a read-twice oblivious BP on 2n Boolean variables (x_1,…,x_n,y_1,…,y_n) with width w and length 4n, layer transition matrices M_j(0),M_j(1)∈{0,1}^{w×w}, and layer-difference matrices

$N_j := M_j(1) - M_j(0)$; for each variable v with its two read-layers (a_v, b_v) form the paired operator $P_v := N_{a_v} N_{b_v} \in R^{w\times w}$, and define the operator-SoS 4-trace gap $\rho(P) := \log_2 \max_{u\neq v} |\langle P_u, P_v\rangle_F| / w^2$, where $\langle A,B\rangle_F = tr(A B^T)$. CONJECTURE: (Upper anchor, trivial via SoS-Cauchy-Schwarz) $\rho(P) \leq 2\cdot\log_2 \text{width}(P) + 2$ for every read-twice oblivious BP P; (Lower bound, non-trivial) for every BLOCKS-ORDERED read-twice oblivious BP P (all x-layers preceding all y-layers in each of the two reads) that computes $IP_2(x,y)=\oplus_i x_i y_i$ with bias $\geq 1/2 + n^{-2}$, $\rho(P) \geq n/16 - 4\cdot\log_2 n$. Equivalently, any such blocks-ordered read-twice BP for IP_2 has width $\geq 2^{n/40}$ for all sufficiently large n; a single small-width blocks-ordered read-twice BP for IP_2 with $\rho(P) < n/16 - 4\cdot\log_2 n$ falsifies the conjecture.

Rationale

The paired operator $P_v = N_{a_v} N_{b_v}$ packages the two reads of variable v into a single non-commutative SoS-style moment, and the Frobenius pairing $\langle P_u,P_v\rangle_F$ is exactly the kind of trace identity that survives Lasserre relaxation at level 2 yet remains computable directly from the BP. For IP_2 the algebraic obstruction that forces exponential read-once size — namely that x and y blocks cannot be summarized in fewer than 2^n states without bilinear cancellation — should reappear in the read-twice setting as a forced spread of mass across many variable-pairs in the Gram matrix of $\{P_v\}$, producing a heavy off-diagonal entry. Because ρ depends on the specific BP layer matrices (not the truth table) it cannot collapse to a function-symmetric invariant, side-stepping the natural-proofs trap that killed the prior Lasserre-rank attempt in this approach.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8f1e5764.py", line 113, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8f1e5764.py", line 69, in run_trial
    N = generate_bp(w, n, seed)
        ^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8f1e5764.py", line 44, in generate_bp
    N = [matrix_subtract(M[j], M[j-1]) if j > 0 else M[j] for j in range(2*n)]
         ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8f1e5764.py", line 28, in matrix_subtra
    w = len(A)
        ^^^^^^
TypeError: object of type 'int' has no len()
```

Judge reasoning

The test crashed with a TypeError in matrix_subtract before producing any data, so neither the upper-anchor nor the IP_2 lower-bound criterion was evaluated. No RESULT line was emitted and the pre-registered support condition cannot be assessed. | next: Fix the matrix representation bug in generate_bp/matrix_subtract (ensure M[j] is a 2D list/array of shape w×w, not an int) and re-run the 540 random instances plus the 4 explicit IP_2 BPs.

Cross-Side Fourier Symmetric-Difference Mass Lower-Bounds DISJ CC_R

- **Verdict**: `BARRIER_HIT`
- **Bridge**: Fourier analysis of Boolean functions (Bonami–Beckner / KKL-style level-weighting on the cube, with bipartite cross-coordinate noise operators rarely used in CC) × Randomized two-party communication complexity of DISJOINTNESS (CC_R) and other partial functions on $N=2^n \times 2^n$ matrices
- **Recorded**: 2026-05-17 03:06 UTC
- **Entry ID**: `fba37ecf9a43`

Statement

For a Boolean matrix $M \in \{0,1\}^{\{2}n \times 2^n\}$ viewed as a function $f\_M : \{0,1\}^{\{2n\}} \to \{0,1\}$ with first n bits the x-input and last n bits the y-input, and for $T \subseteq [2n]$ let $T\_x = T \cap [n]$ and $T\_y = \{i \in [n] : i+n \in T\}$, define $\delta(T) := |T\_x \triangle T\_y|$ and the cross-asymmetry invariant $\Psi(f\_M) := (\Sigma\_T f\_M\_hat(T)^2 \cdot 2^\{\delta(T)\}) / (\Sigma\_T f\_M\_hat(T)^2)$. Conjecture: there exists an absolute constant $c \geq 1/8$ such that for every Boolean matrix M, $CC\_R^\{1/3\}(M) \geq c \cdot \log\_2 \Psi(f\_M)$, and moreover $\Psi(f\_\{DISJ\_n\}) = (7/6)^n$ exactly (so $\log\_2 \Psi = n \cdot \log\_2(7/6) \approx 0.222$ n and $CC\_R(DISJ\_n) = \Omega(n)$ follows). A single Boolean matrix M with a verified protocol of cost B and $\log\_2 \Psi(f\_M) > 8 \cdot B$ refutes the conjecture.

Rationale

Standard Fourier ▯_1 / spectral norm lower bounds for CC are oblivious to the bipartition structure of the input, while CC_R cares essentially about which Fourier modes 'mix' the two parties. Weighting each Fourier mass by $2^\{|T\_x \triangle T\_y|\}$ amplifies modes that genuinely couple x- and y-coordinates asymmetrically — EQ-type matrices live on the diagonal $T\_x = T\_y$ and get $\Psi = 1$, while DISJ's product-over-coordinates structure forces a multiplicative gain of 7/6 per coordinate, giving the right Razborov-type $\Omega(n)$. The hypothesis is therefore a Fourier-analytic 'cross-side' refinement of the standard discrepancy/spectral-norm method, in the spirit of KKL-style coordinate-weighted Fourier inequalities.

Novelty

- Judge: “ over 0 arXiv hits

Empirical Test

- exit code: `-1`, elapsed: `0.00s`

```
(no output)
```

Judge reasoning

Rejected by NATURAL_PROOFS barrier

---

q-Major Cancellation Gap Separates Permanent from Determinant Supports

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Mahonian statistics on the symmetric group (major index, q-factorials, RSK cocharge) × Padded permanent/determinant comparison on shared 0/1 support matrices (GCT det-vs-perm flavor)
- **Recorded**: 2026-05-17 03:32 UTC
- **Entry ID**: `796531c3b7fa`

Statement

For an n x n matrix M in {0,1}^{n x n}, let T_M = {sigma in S_n : M[i,sigma(i)] = 1 for all i} and define the q-major permanent and signed q-major determinant on the support T_M by perm_q(M) := sum_{sigma in T_M} q^{maj(sigma)} and det_q(M) := sum_{sigma in T_M} sign(sigma) * q^{maj(sigma)}, where maj(sigma) is the major index (sum of descent positions). Conjecture: for every n >= 3 and every 0/1 matrix M with |T_M| >= 2, the q=2 cancellation ratio R_2(M) := perm_2(M) / max(1, |det_2(M)|) satisfies R_2(M) >= sqrt(|T_M|) / (2*n). Equivalently, the sign(sigma)-weighted Mahonian sum on any 0/1-support exhibits at least square-root-of-support cancellation against the unsigned version, up to a linear factor in n.

Rationale

Permanent vs determinant in GCT differs precisely by replacing sign(sigma) with 1 inside the S_n-sum; weighting both sums by q^{maj(sigma)} (a Schur-positive Hecke-algebra principal-specialization) refracts that difference through irreducible S_n-character spectra, and at q=2 one already sees [n]_2! = prod(2^k - 1) for perm_n but a much smaller signed sum for det_n. A square-root cancellation lower bound on every common support T_M would certify that determinantal coefficient sequences on permanental supports always look 'random-like' relative to their unsigned partners, giving a Mahonian witness for the per-vs-det gap that is preserved under support-preserving substitutions.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [1503.00822v2] Product Ranks of the 3×3 Determinant and Permanent - [2302.11221v6] A q-analog of certain symmetric functions and one of its specializations - [1302.4419v3] The determinant of the Malliavin matrix and the determinant of the covariance matrix for multiple integrals - [0810.2731v3] Fix-Euler-Mahonian statistics on wreath products - [1101.4332v2] Mahonian Pairs

Empirical Test

- exit code: `1`, elapsed: `1.37s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c5e11f1c.py", line 51, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c5e11f1c.py", line 33, in run_trial
    perm_q = sum(2 ** sum(i for i, x in enumerate(sigma) if sigma[i] > sigma[j]) for sigma in permutation
             ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c5e11f1c.py", line 33, in <genexpr>
    perm_q = sum(2 ** sum(i for i, x in enumerate(sigma) if sigma[i] > sigma[j]) for sigma in permutation
                      ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c5e11f1c.py", line 33, in <genexpr>
    perm_q = sum(2 ** sum(i for i, x in enumerate(sigma) if sigma[i] > sigma[j]) for sigma in permutation
                                                                              ^
NameError: name 'j' is not defined
```

Judge reasoning

The test crashed with a NameError ('j' undefined) in the major-index computation before any trial completed, so no data was produced to evaluate the pre-registered criterion. | next: Fix the maj(sigma) computation to correctly count descent positions (i where sigma[i] > sigma[i+1]) and rerun the 240-trial sweep.

Immanant Positivity Width Bounds Monotone KW Depth for Perfect Matching

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Stembridge–Stanley immanant theory: the λ-immanants imm_λ(S) = Σ_σ χ_λ(σ) Π_i S[i,σ(i)] organise matchings of a 0/1 matrix into Schur–Weyl S_n-isotypic cells (a corner of symmetric-group representation theory with very few papers connecting it to monotone circuit lower bounds outside of GCT speculation) × Monotone Karchmer–Wigderson tree depth d_KW(S) at a 1-input S of the bipartite perfect-matching function PM_n, equivalent by KW to log_2 of the local monotone formula leaf size restricted to matchings inside S vs. small bipartite vertex covers
- **Recorded**: 2026-05-17 04:06 UTC
- **Entry ID**: `15b45d9e4966`

Statement

For 3 ≤ n ≤ 6 and any n×n 0/1 matrix S with perm_n(S) ≥ 1, let Σ(S) := {σ ∈ S_n : S[i,σ(i)] = 1 ∀i} and define the immanant positivity width w(S) := #{λ ⊢ n : imm_λ(S) > 0}. Let d_KW(S) be the minimum depth of any KW protocol that on Alice-input τ ∈ Σ(S) and Bob-input C, an (n−1)-vertex cover of the bipartite complement of some 0-input S′, outputs an edge (i, j) ∈ τ C. Conjecture: d_KW(S) ≥ ⌈log_2 w(S)⌉ for every such S, with equality only when Σ(S) is RSK-shape-monochromatic; in particular w(I_n) = p(n), forcing d_KW(I_n) ≥ ⌈log_2 p(n)⌉ = Θ(√n), a representation-theoretic obstruction in the spirit of GCT's perm/det separation.

Rationale

Each rectangle of a KW protocol pools matchings that all hit a common edge of Bob's cover; we conjecture this pooling is incompatible with carrying more than one irreducible χ_λ-component, so the number of non-vanishing λ-immanants is a Schur-Weyl obstruction to small KW trees. This mirrors the GCT philosophy that perm_n's orbit closure is Schur-rich while det_m's is sign-isotypic: a small substitution-stable Schur-width invariant would translate directly into a perm-vs-det monotone gap. The link of immanants to KW depth is, to our knowledge, untouched in the literature.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [2107.05128v2] Karchmer-Wigderson Games for Hazard-free Computation - [2002.07444v1] Multiparty Karchmer-Wigderson Games and Threshold Circuits - [2003.09602v2] The Number of Perfect Matchings in Möbius Ladders and Prisms - [2304.05285v1] Hook-Shape Immanant Characters from Stanley-Stembridge Characters - [1801.08709v2] Adaptive Lower Bound for Testing Monotonicity on the Line

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_70051383.py", line 178, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_70051383.py", line 132, in run_trial
    d_KW = kw_depth(matrix, covers)
           ^^^^^^^^^^^^^^^^^^^^^^^^
```

```
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_70051383.py", line 104, in kw_depth
    subsets[cover].append(sigma)
    ~~~~~~~^^^^^^^
TypeError: unhashable type: 'set'
```

Judge reasoning

The test crashed with a TypeError (‘unhashable type: set’) in kw_depth before producing any data, so no metrics were computed and the pre-registered support condition cannot be evaluated. | next: Fix the kw_depth implementation by using a hashable key (e.g., frozenset) for cover keys in the subsets dict, then rerun the 360-matrix sweep.

Curto-Fialkow Hankel Defect Separates Read-Twice IP_2 BPs

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Curto-Fialkow truncated moment problem theory (Hankel positivity and flat-extension rank, the real-algebraic-geometry foundation of the Lasserre SoS hierarchy; Curto-Fialkow 1996, Lasserre 2001) applied to layer-conditional empirical moments of branching programs × Read-twice oblivious branching program size for IP_2 on 2n variables in adversarial order x_1,…,x_n,y_1,…,y_n (Borodin-Razborov-Smolensky 1993 open separation)
- **Recorded**: 2026-05-17 04:32 UTC
- **Entry ID**: `9f09553bfd37`

Statement

Let P be a read-twice oblivious BP with state set Q (|Q|=s) computing a Boolean function on N=2n variables, and let q(z,□) denote its state on input z after □ steps. For each state q∈Q and each layer □∈{1,…,2N}, define the centered conditional moment sequence m^{q,□}*k(P) = 2^{-N} Σ_z □[q(z,□)=q]·(IP_2(z)-1/2)^k for k=0,…,⌊N/4⌋, and let H^{q,□}(P) be the (d+1)×(d+1) Hankel matrix with d=⌊N/8⌋ and entries m^{q,□}*{i+j}. Define the positivity defect δ(P) = max_{q,□} #{negative eigenvalues of H^{q,□}(P)}. Conjecture: (i) for every read-twice oblivious BP P of size s, δ(P) ≤ 8·⌈log_2(s+1)⌉; (ii) every read-twice oblivious BP P that exactly computes IP_2 in the adversarial schedule satisfies δ(P) ≥ ⌈√N/2⌉.

Rationale

Curto-Fialkow theory equates degree-2d SoS feasibility with the existence of flat positive extensions of truncated Hankel sequences, so a small BP forces low-rank flat extensions of its layer-conditional empirical moments — a structural constraint never previously exploited for BP lower bounds. IP_2’s bilinear ⊕-structure should force every conditional sub-population reaching a given state at a given layer to retain non-trivial bilinear correlation between the unread x-bits and the moments, obstructing flat positive Hankel extension and producing many negative eigenvalues. The conjectured gap log(s) vs √N would yield s ≥ 2^{Ω(√N)} for read-twice IP_2, attacking the BRS-1993 problem via the SoS-moment side rather than via communication or partition arguments.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
```

```
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8925cd5b.py", line 155, in <module>
    result = run_trial(int(seed))
             ^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8925cd5b.py", line 104, in run_trial
    P, Q, N = generate_random_bp(n, w, seed)
              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8925cd5b.py", line 64, in generate_rand
    P[(q, l)][k] += 1
    ~~~~~~~~~^^^
TypeError: list indices must be integers or slices, not float
```

Judge reasoning

The test crashed with a TypeError in generate_random_bp before producing any data, so neither the support nor falsification condition could be evaluated. | next: Fix the indexing bug in generate_random_bp (cast the index k to int) and rerun the pre-registered sweep.

Hodge-Cheeger Product Lower-Bounds Tseitin Resolution Length

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Combinatorial Hodge theory on graphs (pairing of harmonic 1-cochain dimension b_1 with vertex-Laplacian algebraic connectivity, in the Lim 2020 / Jiang-Lim-Yao-Ye framework used in TDA but essentially unused in proof complexity) × Tree-like Resolution refutation length of Tseitin XOR formulas T(G,omega) on bounded-degree graphs G with odd charge omega
- **Recorded**: 2026-05-17 05:07 UTC
- **Entry ID**: `b60e3f3f26e1`

Statement

Let G=(V,E) be a connected graph with max degree d_max <= 6 and |V|=n, and let omega:V->F_2 satisfy sum_v omega_v = 1 (mod 2). Define b_1(G) := |E|-|V|+1 (the first combinatorial Betti number, i.e. cycle-space dimension over F_2), let lambda_2(L_G) denote the second-smallest eigenvalue of the combinatorial vertex Laplacian L_G=D-A, and set the Hodge-Cheeger invariant nu(G) := lambda_2(L_G) * b_1(G) / d_max. Conjecture: there is an absolute constant c >= 0.5 such that every tree-like Resolution refutation of T(G,omega) has length at least 2^{c*nu(G)}; a single (G,omega) with nu(G) >= 5 and a Resolution refutation of length < 2^{0.5*nu(G)} falsifies the conjecture.

Rationale

Tseitin hardness needs two ingredients: enough independent cycles to encode unsatisfiable XOR obstructions (measured by b_1) and enough spectral mixing to prevent localized cancellation (measured by lambda_2); combinatorial Hodge theory packages exactly this pairing via the orthogonal decomposition of edge flows into gradients and harmonic 1-cochains. Bridges/paths zero out b_1; bottlenecked graphs zero out lambda_2; only graphs that are simultaneously cycle-rich and spectrally expanding (the BSW regime) yield large nu, matching the empirical hard-Tseitin profile. The framing is novel for proof complexity (<5 papers connect Hodge theory to Resolution) yet entirely poly-time computable via a single numpy eigendecomposition.

Novelty

- Judge: `NOVEL` over 10 arXiv hits

Top hits consulted: - [2209.05839v3] On bounded depth proofs for Tseitin formulas on the grid; revisited - [2103.09609v1] Characterizing Tseitin-formulas with short regular resolution refutations

- [0906.0693v3] An improved lower bound on the counterfeit coins problem - [1301.2045v3] Integral-valued polynomials over the set of algebraic integers of bounded degree - [1606.05050v1] Proof Complexity Lower Bounds from Algebraic Circuit Complexity

Empirical Test

- exit code: `124`, elapsed: `240.10s`

```
TIMEOUT after 240s
```

Judge reasoning

The test timed out after 240s (returncode 124) before producing any RESULT line or data, so the pre-registered support condition cannot be evaluated. | next: Reduce instance count or graph sizes (e.g., shrink to 3 categories × 3 sizes × 10 seeds) and/or impose a per-instance Resolution-search timeout so the suite completes within the wall-clock budget.

Sandpile Group Order Lower-Bounds Monotone KW Depth

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Abelian sandpile / chip-firing group theory (Dhar 1990, Bjorner-Lovasz-Shor 1991; the critical/sandpile group K(G) = coker of the reduced graph Laplacian, |K(G)| = number of spanning trees by Kirchhoff). This corner of algebraic combinatorics / statistical mechanics has essentially no presence in monotone circuit lower-bound literature. × Monotone Karchmer-Wigderson protocol depth D_KW(f) = log_2 of minimum monotone formula leaf size for a monotone Boolean function f, accessed via the bipartite KW relation (minterm, maxterm) -> intersection coordinate.
- **Recorded**: 2026-05-17 05:45 UTC
- **Entry ID**: `f75d64a69735`

Statement

For every nonconstant monotone Boolean f:{0,1}^n -> {0,1} let M(f) and K(f) be its minterm and maxterm antichains (as subsets of [n]). Form the bipartite multigraph G(f) on vertex set M(f) ⊔ K(f) with an edge of multiplicity |m∩k| between each m∈M(f) and k∈K(f), add a ground vertex g joined by a single edge to every k∈K(f), let tau(f) = number of spanning trees of (G(f)∪{g}) computed via the matrix-tree theorem on the reduced Laplacian, and put sigma(f) = log_2(1+tau(f)). Then sigma(f) ≤ D_KW(f) · (|M(f)|+|K(f)|+1) · log_2(n+2), with the constant 1 in front being the conjectured tight prefactor. A single monotone f exhibiting sigma(f) strictly above this bound (using any verified upper bound on D_KW(f) from an explicit protocol) falsifies the conjecture.

Rationale

The sandpile group of the KW bipartite intersection multigraph counts independent 'cycles of certificates' that any KW protocol must resolve; each protocol step (a coordinate query at an internal node) can collapse at most O((|M|+|K|) log n) such independent cycles by a Cheeger-style averaging on the Laplacian, suggesting depth lower-bounds via log|K(G)|. Because tau(f) is determined by the BOTH-sided minterm/maxterm relation rather than a single-number truth-table statistic, the invariant is relational and exponential-time in the truth table, so it sidesteps the Razborov-Rudich natural-proofs barrier and the ring-only algebrization trap.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [2107.05128v2] Karchmer-Wigderson Games for Hazard-free Computation - [1012.2478v2] Packing spanning trees and the k-tree protocol - [0904.3087v4] Distributed Maintenance of Anytime Available Spanning Trees in Dynamic Networks - [1606.05050v1] Proof Complexity Lower Bounds from Algebraic Circuit Complexity - [1904.05483v2] Parallels Between Phase Transitions and Circuit Complexity?

Empirical Test

- exit code: `124`, elapsed: `240.10s`

```
TIMEOUT after 240s
```

Judge reasoning

The test timed out after 240s (returncode 124) without producing a RESULT line, so the pre-registered support condition cannot be evaluated and the critic challenge stands. | next: Optimize the test harness (e.g., cache D_KW upper bounds, reduce seed count or cap n at 6) so all 35 instances complete within the time budget.

Conway Temperature Bounds Tree-Resolution Width for 3-SAT

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Conway-Berlekamp combinatorial game theory: left/right stops and temperature of finite alternating scoring games (Conway 1976; Berlekamp-Wolfe 1994; Siegel 2013). This corner of CGT has essentially no presence in proof complexity outside Beck's positional-game tradition; a focused arXiv search for 'Conway temperature' AND ('resolution' OR 'CNF' OR 'proof complexity') returns no hits. × Tree-like Resolution refutation width w*(F) and size s*(F) for unsatisfiable 3-CNFs, viewed in the Cook-Reckhow-Krajicek bounded-arithmetic framework where ¬F is a Σ^b_0 statement whose $V^{0/S}$1_2 witnessing complexity is tied by Ben-Sasson–Wigderson to Resolution width and by Pudlak–Impagliazzo to the Prover–Adversary game.
- **Recorded**: 2026-05-17 06:17 UTC
- **Entry ID**: `18df83a46e71`

Statement

For unsat 3-CNF F on n ≤ 14 variables, define the alternating integer-scoring game G(F): players A (Maximizer) and B (Minimizer) alternate; on each turn the active player picks one unassigned variable and freely assigns a value; when all variables are assigned the score equals the number of falsified clauses. Compute the Conway left stop L(F) (Maximizer to move) and right stop R(F) (Minimizer to move) by memoized minimax on the full 2^n game tree, and let t(F) := (L(F)−R(F))/2 be the (degree-0) Conway temperature. Conjecture: (i) t(F) ≤ C_1 · w*(F) for every unsat 3-CNF F, where w*(F) is the tree-Resolution refutation width, and (ii) t(F) ≥ C_2 · n for Tseitin formulas T(G,ω) on 3-regular expander graphs G with v ≤ 10 vertices; a single instance violating either bound refutes the conjecture (defaults C_1 = 5, C_2 = 0.3).

Rationale

Tree-Resolution width is controlled by adversarial variable choice in the Prover-Adversary game, while the Conway temperature measures opposite-sign oscillation in optimal play of a scoring game — a structurally distinct invariant never (to our knowledge) computed on SAT-derived games. Tseitin formulas on expanders have Ω(n) resolution width by parity-flow, and the parity structure should also force balanced left/right moves at every vertex-clause group, yielding high temperature. If the bridge holds it gives a witnessing-style certificate of width inside the bounded-arithmetic Σ^b_0 Cook-Reckhow framework.

Novelty

- Judge: `NOVEL` over 6 arXiv hits

Top hits consulted: - [1709.03479v2] Conway's potential function via the Gassner representation - [2203.14669v1] Dynamic Structure in Four-strategy Game: Theory and Experiment - [2207.14140v1] Playing a 2D Game Indefinitely using NEAT and Reinforcement Learning - [2501.08313v1] MiniMax-01: Scaling Foundation Models with Lightning Attention - [2506.13585v1] MiniMax-M1: Scaling Test-Time Compute Efficiently with Lightning Attention

Empirical Test

- exit code: `0`, elapsed: `4.20s`


```
: 1, 'conjecture_holds': False, 'counterexample': 'Random 3-CNF with n=10 violated t(F) ≤ 5·w*(F): t=6
TRIAL: {'metric_name': 'temperature_to_width_ratio', 'metric_value': 0, 'instances_tested': 2, 'conj
CNF with n=10 violated t(F) ≤ 5·w*(F): t=6.5, w*=True'}
TRIAL: {'metric_name': 'temperature_to_width_ratio', 'metric_value': 0, 'instances_tested': 1, 'conj
CNF with n=10 violated t(F) ≤ 5·w*(F): t=6.5, w*=True'}
TRIAL: {'metric_name': 'temperature_to_width_ratio', 'metric_value': 0, 'instances_tested': 2, 'conj
CNF with n=10 violated t(F) ≤ 5·w*(F): t=6.5, w*=True'}
TRIAL: {'metric_name': 'temperature_to_width_ratio', 'metric_value': 0, 'instances_tested': 2, 'conj
CNF with n=14 violated t(F) ≤ 5·w*(F): t=9.5, w*=True'}
TRIAL: {'metric_name': 'temperature_to_width_ratio', 'metric_value': 0, 'instances_tested': 2, 'conj
CNF with n=10 violated t(F) ≤ 5·w*(F): t=6.0, w*=True'}
TRIAL: {'metric_name': 'temperature_to_width_ratio', 'metric_value': 0, 'instances_tested': 1, 'conj
CNF with n=10 violated t(F) ≤ 5·w*(F): t=6.0, w*=True'}
TRIAL: {'metric_name': 'temperature_to_width_ratio', 'metric_value': 0, 'instances_tested': 1, 'conj
CNF with n=8 violated t(F) ≤ 5·w*(F): t=5.5, w*=True'}
TRIAL: {'metric_name': 'temperature_to_width_ratio', 'metric_value': 0, 'instances_tested': 1, 'conj
CNF with n=8 violated t(F) ≤ 5·w*(F): t=6.0, w*=True'}
TRIAL: {'metric_name': 'temperature_to_width_ratio', 'metric_value': 0, 'instances_tested': 1, 'conj
CNF with n=8 violated t(F) ≤ 5·w*(F): t=5.5, w*=True'}
TRIAL: {'metric_name': 'temperatu
```


Judge reasoning

The 'w' *field is reported as the boolean True rather than an integer tree-Resolution width, so the comparison t(F) ≤ 5·w*(F) is being evaluated against a coerced value of 1 rather than the actual refutation width. The reported falsifications are artifacts of a broken width oracle, not genuine counterexamples. | next: Fix the tree-Resolution width computation to return an integer (exact tree-DPLL depth/width), re-run the 30-seed protocol, and additionally verify the Tseitin lower-bound clause se

---

Hochster Regularity of Variable Co-Occurrence Graph Lower-Bounds Resolution Width

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Castelnuovo–Mumford regularity of squarefree monomial edge ideals via Hochster's combinatorial formula on independence complexes (Hochster 1977; Bruns–Herzog 1993; Katzman 2006; Woodroofe 2014) — a corner of computational commutative algebra with essentially no proof-complexity-side publications (arXiv search 'edge ideal regularity' AND ('resolution width' OR 'proof complexity') returns no hits) × Resolution refutation width w*(F) for unsatisfiable 3-CNFs, viewed as a Ben-Sasson–Wigderson and Krajícek-style stepping stone toward size lower bounds for tree-like Resolution and Extended Frege (via the depth-O(1) Frege simulation of Resolution)

- **Recorded**: 2026-05-17 06:44 UTC
- **Entry ID**: `e0a06bc1af46`

Statement

For an unsatisfiable 3-CNF F on $n \leq 10$ variables, build the variable co-occurrence graph G(F) with V = {x_1,…,x_n} and edges {x_i,x_j} whenever x_i and x_j co-appear in at least one clause of F. Define reg(F) := max{ j ≥ 0 : ∃ W ⊆ V with reduced rational homology H̃*{|W|−j−1}(Δ*{G(F)[W]}; Q) ≠ 0 }, where Δ_{G[W]} is the independence complex of the induced subgraph G[W] (Hochster's formula for reg(Q[x]/I_{G(F)})). Conjecture: for every unsatisfiable 3-CNF F, the minimum Resolution refutation width satisfies w*(F) ≥ reg(F) + 2; equivalently, a single F with w*(F) < reg(F) + 2 falsifies it.

Rationale

Hochster's formula recasts an algebraic invariant of the edge ideal as the maximum homological complexity over induced subgraphs, identifying variable clusters whose mutual logical entanglement is topologically nontrivial; intuitively, no narrow refutation can simultaneously break such a cluster, forcing high resolution width. The bridge from edge-ideal regularity to proof complexity is essentially unstudied, and unlike single-number truth-table invariants (calibration trap (j)), reg(F) is a *relational* invariant of the CNF graph computed by quantifying over induced substructures, sidestepping the Razborov–Rudich largeness criterion that defeats truth-table approaches. If valid, BSW gives size ≥ $\exp(\Omega(\mathrm{reg}(F)^2/n))$, and via Krajícek's depth-d Frege simulation this propagates to nontrivial EF line-count certificates.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.15s`

```
TRIAL: {'metric_name': 'w*(F) - reg(F)', 'metric_value': 0, 'instances_tested': 1, 'conjecture_holds'

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0d752578.py", line 149, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0d752578.py", line 133, in run_trial
    w = resolution_width(F, n)
        ^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0d752578.py", line 108, in resolution_w
   if any((var, True) in c1 and (var, False) in c2) or any((var, False) in c1 and (var, True) in c2):
      ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
TypeError: 'bool' object is not iterable
```

Judge reasoning

The test crashed with a TypeError in resolution_width (misuse of any() on a boolean expression), producing only 1 trial instead of the pre-registered 90 seeds. No valid data was collected to evaluate the support condition. | next: Fix the resolution_width function (the any() calls wrap non-iterable boolean expressions) and rerun the full 90-seed sweep across n∈{6,8,10}.

Ihara Zeta Entropy Predicts Tseitin DPLL Tree Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Ihara zeta function of finite graphs (Hashimoto–Stark–Terras combinatorial graph-zeta theory rooted in number-theoretic L-functions; essentially no papers connect Ihara zeta values to proof complexity, distinct from non-backtracking spectral approaches) × Tree-like Resolution / DPLL refutation tree size for Tseitin XOR formulas T(G,ω) on 3-regular graphs (Urquhart–Ben-Sasson–Razborov–Atserias regime)
- **Recorded**: 2026-05-17 07:23 UTC
- **Entry ID**: `3ba5471eff37`

Statement

Let G be a connected 3-regular graph on n vertices with adjacency matrix A, and let ω:V→{0,1} have Σω odd, so T(G,ω) is unsatisfiable. Define the Ihara entropy H(G) := (1/n)·ln det((11/9)·I − A/3), which is well-defined and positive because (11/3)·I − A has eigenvalues in [2/3, 20/3] for any 3-regular G. We conjecture that with pre-registered absolute constants c_1 = 1/20 and c_2 = 20, every connected 3-regular G with odd ω satisfies c_1·n·H(G) − 1 ≤ $\log_2$(S_DPLL(T(G,ω))) ≤ c_2·n·H(G) + $\log_2$(n) + 1, where S_DPLL is the leaf count of the canonical DPLL refutation tree using unit propagation plus first-unassigned-variable branching. A single (G,ω) violating either inequality refutes the conjecture.

Rationale

The Ihara zeta function ζ_G(u) = ∏_p (1−u^{|p|})^{−1} over prime non-backtracking cycles is encoded by det(I − uA + u^2(D−I)); at u = 1/3 this determinant aggregates how cycles of all lengths contribute multiplicatively, and each cycle of G corresponds in T(G,ω) to a closed XOR constraint that DPLL must locally certify, so per-vertex cycle entropy should track per-vertex branching entropy. The evaluation point u = 1/3 sits strictly inside the radius of convergence for 3-regular graphs, giving a numerically stable, integer-matrix invariant whose growth on Ramanujan-type expanders matches Urquhart's exponential refutation-size lower bound. This bridges number-theoretic graph zetas to proof complexity in a regime where, to our knowledge, fewer than five papers have looked.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.01s`

```
TIMEOUT after 240s
```

Judge reasoning

The test timed out after 240s (returncode=124) without producing a RESULT line or any instance data, so neither the bound inequalities nor the Spearman ρ threshold could be evaluated. | next: Reduce the instance budget (e.g. fewer seeds or cap n at 12) and/or replace the exact DPLL leaf count with an iterative bounded search so the 150-instance sweep completes within the timeout.

Signed-Support L_inf Discrepancy Bounds Bottom Fan-In for AC0 PARITY

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Combinatorial discrepancy theory (Beck-Fiala, Spencer six-standard-deviations, Chazelle-Matousek hereditary discrepancy of ±1 incidence matrices) × AC^0 depth-d circuit size and bottom fan-in for PARITY_n, in the Hastad / Yao switching-lemma regime

- **Recorded**: 2026-05-17 08:06 UTC
- **Entry ID**: `2f193a8b2209`

Statement

For an AC^0 circuit C on n inputs with bottom-level AND/OR gates B_1,...,B_m, define the signed support matrix M(C) in {-1,0,+1}^{m x n} by M_ij = +1 if x_j appears positively in B_i, -1 if negatively, and 0 if absent, and let w(C) = max_i ||M_i||*0 be the maximum bottom fan-in. Define the L_infty discrepancy psi(C) := min*{chi in {-1,+1}^n} max_{i<=m} | sum_j M_ij chi_j |. CONJECTURE: there is an absolute constant c >= 1/4 such that for every AC^0 circuit C of any constant depth that computes PARITY_n, psi(C) >= c * w(C); a single PARITY-computing AC^0 instance with psi(C) < (1/4) * w(C) refutes it.

Rationale

PARITY forces every bottom-gate sign pattern to participate in a covering of (n-1)-parity-correct assignments, so the rows of M(C) must span a Hadamard-like sign system that admits no balanced coloring whose row-sums all fall below the typical sqrt(w) scale; this is a structural cancellation obstruction distinct from random restrictions. The invariant is exactly S_n x Z_2^n equivariant (variable relabeling and literal-polarity flips both preserve psi and w), satisfying the focus-mode naturality demand. If correct, the bound psi >= c*w replaces the switching-lemma probability calculation by a deterministic discrepancy certificate and is sharp on the canonical CNF/DNF (where psi ~ n = w).

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.10s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out after 240s (returncode 124) before producing any RESULT line or data; the pre-registered support condition cannot be evaluated. | next: Reduce search space (e.g., restrict to n in {4,6,8} and fewer seeds) or optimize psi computation via branch-and-bound/SDP relaxation to obtain data within the time budget.

2-Sylow Rank of Critical Group Bounds Tseitin DPLL Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Algebraic combinatorics of the critical (sandpile) group K(G): specifically the 2-Sylow rank r_2(G)=dim_{F_2} ker(L̃_G mod 2) of the reduced graph Laplacian, in the Bjorner–Lovasz–Shor / Wood–Mészáros tradition on cokernels of random integer matrices; distinct from the well-studied invariant |K(G)|=τ(G) and essentially absent from Tseitin proof-complexity literature × Tree-like Resolution / DPLL refutation size for Tseitin XOR formulas T(G,ω) on 3-regular graphs (Urquhart–Ben-Sasson–Razborov regime), via the cycle-space-mod-2 mechanism underlying expander Tseitin lower bounds
- **Recorded**: 2026-05-17 08:50 UTC
- **Entry ID**: `eb9e15c9c9db`

Statement

For every connected 3-regular graph G on n vertices and every charge ω:V→F_2 with Σω≡1 (mod 2), let L̃*G be the reduced graph Laplacian (Laplacian with one row/column removed) and define the 2-Sylow rank r_2(G):=(n−1)−rank*{F_2}(L̃_G mod 2). Let N_DPLL(T(G,ω)) be the number of nodes in the canonical DPLL refutation tree of the Tseitin CNF (with unit propagation, branch on lowest-index unassigned variable). Conjecture: log_2 N_DPLL(T(G,ω)) ≥ (1/4)·(n−1−r_2(G)) for every such (G,ω); equivalently, a single (G,ω) with log_2 N_DPLL < (1/4)(n−1−r_2(G)) falsifies it.

Rationale

Tseitin refutation hardness is governed by the cycle-space mod 2 of G, and r_2(G) precisely captures how many independent F_2-degenerate directions the Laplacian pairing has on that cycle space; thus low r_2(G) (full F_2-rigidity) should force DPLL to traverse all cycle-space cosets while high r_2 opens algebraic short-cuts. This invariant is graph-structural (not a truth-table predicate), bypassing natural proofs, and is computed by elementary F_2 linear algebra rather than ring extensions, so it does not algebrize. It connects an under-utilised corner of algebraic combinatorics (2-part of Jacobian cokernel distributions) to the established expander-Tseitin lower-bound machinery.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.10s`

```
TIMEOUT after 240s
```

Judge reasoning

The test timed out after 240s (returncode 124) without producing a RESULT line or any instance data, so the pre-registered support condition over 150 instances cannot be evaluated. | next: Re-run with a longer timeout and/or reduce the size range (e.g., n∈{6,8,10,12}) and cache DPLL counts to obtain complete data on all 150 instances.

Stable-Rank of Layer-Symbol Matrix Separates Read-Twice BPs for IP_2

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Forster-style stable-rank / numerical-rank theory (Forster 2002; Linial–Shraibman 2009) — a corner of sign-rank / discrepancy-based lower-bound machinery transplanted from communication matrices to the operator-valued symbol bundle of a branching program; this bridge has essentially no presence in BP literature outside the unrelated Nechiporuk rank tradition. × Read-twice oblivious branching program size for IP_2 on 2n adversarial-ordered variables (Borodin–Razborov–Smolensky 1993 open separation), in the meta-complexity sense of seeking a 'compressibility witness' ρ(P) with ρ(P)=O(log size(P)) — a poly-time-computable structural invariant of the BP, NOT of its truth-table.
- **Recorded**: 2026-05-17 09:02 UTC
- **Entry ID**: `46c3da6284c4`

Statement

For a read-twice oblivious BP P of size s on 2n variables, with layers t=1..4n and transition matrices M_t(b)∈{0,1}^{s×s} for b∈{0,1}, form the layer-symbol matrix S(P) := [Δ_1 | Δ_2 | … | Δ_{4n}]

$\in \mathbb{R}^{s \times 4n\cdot s}$ where $\Delta_t := M_t(1) - M_t(0)$. Define the stable-rank invariant $\rho(P) := \|S(P)\|_F^2 / \|S(P)\|_{op}^2$. Conjecture: (i) $\rho(P) \leq 2\cdot s$ for every read-twice BP P (numerical-rank anchor: this follows from rank $\leq$ s but we conjecture the tighter constant 2 holds for all syntactic BPs); (ii) for every read-twice BP P with output correlation $\langle f_P, IP_{2\_n}\rangle \geq \varepsilon$ over uniform 2n-bit inputs, $\rho(P) \geq c \cdot \varepsilon^2 \cdot 2^{n/2}$ for an absolute constant $c > 0$. A single instance with $\langle f_P, IP_2\rangle \geq \varepsilon$ and $\rho(P) < c\cdot\varepsilon^2\cdot 2^{n/2}$, or any read-twice BP with $\rho(P) > 2s$, refutes it.

Rationale

Each Δ_t records how flipping one read at layer t shifts the BP's state operator; for a small BP these shifts share an s-dimensional column space, capping the stable rank at O(s). IP_2's adversarial pairing of x_i with y_i forces the symbol bundle to encode $2^{\Theta(n)}$ mutually-orthogonal cross-variable shift directions (after the middle cut, each pair (x_i,y_i) contributes a Hadamard-flavoured outer product), so the Frobenius mass spreads across exponentially many comparable singular values, making $\|S\|_F^2/\|S\|_{op}^2$ grow geometrically with n. The setup avoids natural-proofs (ρ is a structural property of P, not its truth-table), algebrization (singular values are analytic, not ring-arithmetic), and relativization (no oracle).

Novelty

- Judge: `NOVEL` over 6 arXiv hits

Top hits consulted: - [0710.5926v2] Mod 2 cohomology of 2-local finite groups of low rank - [1503.07648v2] Sign rank versus VC dimension - [1708.02054v1] Pseudorandom Bits for Oblivious Branching Programs - [0809.2093v2] An approximation algorithm for approximation rank - [1706.02207v3] On The Communication Complexity of High-Dimensional Permutations

Empirical Test

- exit code: `0`, elapsed: `0.08s`


```
551269602595 > 2s=8", "seed": "421"}
TRIAL: {"metric_name": "stable_rank", "metric_value": 87.00000000000001, "metric_std": 0.0, "instanc
TRIAL: {"metric_name": "stable_rank", "metric_value": 80.24174837076876, "metric_std": 0.0, "instanc
TRIAL: {"metric_name": "stable_rank", "metric_value": 39.25023292246046, "metric_std": 0.0, "instanc
TRIAL: {"metric_name": "stable_rank", "metric_value": 47.12182530062137, "metric_std": 0.0, "instanc
TRIAL: {"metric_name": "stable_rank", "metric_value": 65.81197395850668, "metric_std": 0.0, "instanc
TRIAL: {"metric_name": "stable_rank", "metric_value": 0.0, "metric_std": 0.0, "instances_tested": 0,
TRIAL: {"metric_name": "stable_rank", "metric_value": 36.561209216717984, "metric_std": 0.0, "instan
TRIAL: {"metric_name": "stable_rank", "metric_value": 28.50532667271179, "metric_std": 0.0, "instanc
TRIAL: {"metric_name": "stable_rank", "metric_value": 59.148813421486146, "metric_std": 0.0, "instan
```


Judge reasoning

The critic correctly notes that $\rho(P)=\|S\|_F^2/\|S\|_{op}^2$ is mathematically bounded by rank(S) ≤ s, so reported values like 26–129 with 2s=8 are impossible and indicate a metric-implementation bug (likely summing per-layer ratios instead of computing the stable rank of the concatenated matrix). The pre-registered support condition is therefore not validly evaluated, and the 'falsification' cannot be trusted. | next: Fix the ρ computation to actually compute $\|S\|_F^2/\|S\|_{op}^2$ on the concatenated S∈ℝ^{s×4n

---

TP_2 Minor-Sign Defect of Spectral Pseudo-Moment Matrix Bounds Max-CUT Ratio

- **Verdict**: `INCONCLUSIVE`

- **Bridge**: Karlin–Edrei–Schoenberg total positivity / Pólya frequency theory (TP_k matrices, sign-regularity of 2x2 minors) × Degree-2 SOS pseudo-moment matrix for Max-CUT on graphs (Goemans–Williamson regime), with a lifting interpretation via gadget composition
- **Recorded**: 2026-05-17 09:29 UTC
- **Entry ID**: `16809e692923`

Statement

For a connected graph G=(V,E) on n vertices with combinatorial Laplacian L=D-A, let v_2,v_3∈R^n be the second and third eigenvectors of L (unit-norm, breaking ties by lexicographic sign). Define the rank-2 spectral pseudo-moment matrix M_G[i,j]:=(v_2[i]v_2[j]+v_3[i]v_3[j])/Z, where Z:=max_{a,b}|v_2[a]v_2[b]+v_3[a]v_3[b]|, and the TP_2 defect δ(G):=|{(i<j,k<l):det M_G[{i,k},{j,l}]<-1/$n^{3\}|/\mathrm{binom}(n,2)}$2. Conjecture: there exist absolute constants c_1=0.5 and c_2=0.10 such that (i) every graph with max-CUT(G)/|E|≥0.879 satisfies δ(G)≤c_1·n^{-1/2}, and (ii) every graph on n≥12 vertices with max-CUT(G)/|E|≤0.55 satisfies δ(G)≥c_2. A single graph violating either inequality refutes the conjecture.

Rationale

Pólya frequency / TP_2 matrices are exactly the moment matrices of log-concave one-parameter families; SOS-tight max-CUT instances (where degree-2 SOS already certifies a near-optimal cut) should correspond to log-concave 1D rounding directions and thus preserve TP_2 in the leading spectral block, whereas integrality-gap instances spread mass in a way that flips 2x2 minor signs. Because TP_2 is closed under coordinatewise polynomial gadget tensoring (Karlin 1968, Ch. 3), δ behaves as a lifting-friendly invariant: gadget composition can only grow δ by a constant factor, matching the 'preserved under polynomial reductions' requirement on the target property P. The bridge between Pólya frequency theory and SOS Max-CUT lower bounds appears to have under five papers (TP_k has been used in stable-polynomial proofs of Kadison–Singer but not in SOS degree-lower-bound machinery).

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `3.25s`

```
TRIAL: {'metric_name': 'TP_2 defect', 'metric_value': 0.0004911287328869746, 'instances_tested': 8,

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_41095b21.py", line 253, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_41095b21.py", line 216, in run_trial
    delta = tp2_defect(graph)
            ^^^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_41095b21.py", line 186, in tp2_defect
    M = spectral_pseudo_moment_matrix(graph)
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_41095b21.py", line 174, in spectral_pse
    eigenvalues, eigenvectors = matrix_eigen(L)
                                ^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_41095b21.py", line 80, in matrix_eigen
    Q, R = qr_decomposition(A)
           ^^^^^^^^^^^^^^^^^^^
```

```
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_41095b21.py", line 96, in qr_decomposit
    Q[i][k] = A[i][k] / R[k][k]
              ~~~~~~~~^~~~~~~~
ZeroDivisionError: float division by zero
```

Judge reasoning

The test crashed with a ZeroDivisionError in the custom QR decomposition after only 8 instances, well short of the pre-registered ~600 instance protocol. Although a putative low-cut counterexample was reported, the run terminated abnormally and the support condition cannot be evaluated. | next: Replace the hand-rolled QR with numpy.linalg.eigh (or guard against zero pivots) and rerun the full 4×30×5 protocol before judging the conjecture.

Gromov 4-Point δ of XOR-Lifted Row Metric Bounds DT Depth

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Coarse geometry / Gromov 4-point δ-hyperbolicity of finite metric spaces (Gromov 1987; Bridson–Haefliger 1999; Coornaert 2014) — a relational metric invariant from geometric group theory with essentially no presence in communication complexity (search 'Gromov hyperbolicity' AND 'communication complexity' returns no direct hits; <5 papers in adjacent areas) × Deterministic two-party communication complexity of XOR-gadget-lifted Boolean functions, in the Raz–McKenzie / Göös–Pitassi–Watson / Hatami–Hossein-Pitassi query-to-communication lifting framework, accessed through the bipartite row metric of the lifted communication matrix M_{f∘XOR_2}
- **Recorded**: 2026-05-17 10:01 UTC
- **Entry ID**: `5ee62d63ba3c`

Statement

Let f:{0,1}^n→{0,1} have deterministic decision-tree depth d=DT(f), let g=XOR_2 (Alice has x_i∈{0,1}, Bob has y_i∈{0,1}, g(x_i,y_i)=x_i⊕y_i), and let M∈{0,1}$^{\{2}$n×2^n} be the lifted matrix M[x,y]=f(x⊕y). View rows of M as points in ({0,1}$^{\{2}$n},d_H). For each 4-tuple of distinct rows (a,b,c,d) compute the three sums S_1=d_H(a,b)+d_H(c,d), S_2=d_H(a,c)+d_H(b,d), S_3=d_H(a,d)+d_H(b,c), sort them S_{(1)}≥S_{(2)}≥S_{(3)}, and set δ(M)=max over 4-tuples of (S_{(1)}−S_{(2)})/2; let δ(M)=δ(M)/diam(M). Conjecture: for every non-constant f, δ(M_{f∘XOR_2}) ≤ 1 − DT(f)/(n+1), i.e. higher DT-depth forces strictly more 4-point-treelike row geometry; a single f with δ(M) > 1 − DT(f)/(n+1) refutes it.

Rationale

Lifting theorems claim DT structure of f survives into the gadget-amplified bipartite matrix; if so, the inherited combinatorial structure should constrain coarse-geometric quantities of the row Hamming cloud, since high-DT functions composed with XOR_2 build hierarchical query trees that the Gromov 4-point inequality is precisely designed to detect. Crucially δ is a 4-point relational invariant — not a single-number truth-table functional — so it sidesteps the Razborov–Rudich largeness obstruction, and uses only Hamming distances (no ring/polynomial structure), so it should not algebrize. If the bound holds, it gives a coarse-geometric proxy for the lifted communication lower bound via embedding-distortion arguments à la Linial–London–Rabinovich.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f5a45145.py", line 97, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f5a45145.py", line 69, in run_trial
    s3 = distances[sum(range(a+1, d+1))] + distances[sum(range(b+1, c+1))]
         ~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^
IndexError: list index out of range
```

Judge reasoning

The test crashed with an IndexError before producing any data, so neither the support condition nor a counterexample was established. Per the rules, a crash yields INCONCLUSIVE. | next: Fix the indexing bug in the 4-tuple distance lookup (use a proper pair-to-index mapping like distances[(i,j)] or itertools.combinations indexing) and rerun the pre-registered sweep.

Median Half-Subgraph Separator Lower-Bounds Tseitin DPLL Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Separation profile of geometric group theory (Benjamini–Schramm–Timár 2012; Hume–Mackay–Tessera 2020; Hume 2017): the coarse-geometric invariant sep_G(k) = sup over k-vertex subgraphs H of min balanced vertex separator of H, used to distinguish quasi-isometry classes of finitely generated groups. This corner of GGT has essentially no presence in proof complexity — an arXiv search 'separation profile' AND ('resolution' OR 'Tseitin' OR 'proof complexity') returns no hits; the closest neighbours are asymptotic-dimension papers that never reach proof systems. × Tree-like Resolution / DPLL refutation tree size for Tseitin XOR formulas T(G,ω) on 3-regular graphs, in the Urquhart–Ben-Sasson–Wigderson–Krajíček bounded-arithmetic regime where ¬T(G,ω) is a Σ^b_0 statement and DPLL size lower-bounds V^0-witnessing complexity.
- **Recorded**: 2026-05-17 10:32 UTC
- **Entry ID**: `118fd4951a23`

Statement

For a connected 3-regular graph G on m vertices and odd charge ω: ν_30(G) := median over 30 uniformly random vertex subsets U ⊂ V(G) of size k=⌊m/2⌋ of the value β(G[U]), where β(H) is the size of a minimum vertex separator (A,V(H)∖(A∪S),S) of G[U] with |A|,|V(H)∖(A∪S)| ≥ ⌊|V(H)|/3⌋, and β(H):=0 if G[U] is disconnected. Conjecture: (i) every odd-charge Tseitin formula T(G,ω) requires DPLL tree size s*(T(G,ω)) ≥ 2^{0.1·ν_30(G)}; (ii) for every 3-regular G admitting a vertex partition (V_1,V_2) with |V_1|,|V_2| ≥ m/3 and |E(V_1,V_2)| ≤ 2 (e.g., barbells, prism-pairs joined by a bridge or two edges), ν_30(G) ≤ 2. A single 3-regular instance with log_2(s*) < 0.1·ν_30(G), or a small-cut graph in class (ii) with ν_30(G) > 2, refutes the conjecture.

Rationale

The Ben-Sasson–Wigderson width method bounds Tseitin Resolution length via expansion of G *as a whole*; but bounded arithmetic V^0-witnessing of unsatisfiable Σ^b_0-instances should react to *every scale* of expansion, not just the global one. The separation profile evaluated at half-size captures exactly this: it is large iff *typical* half-subgraphs of G are themselves robustly connected, which is the structural data DPLL must repeatedly re-encounter on any branch that fixes half the variables. The link to Cook-Reckhow-Krajíček: ν_30 is a low-complexity (NC-computable up to sampling) Σ^b_0-definable property of G, so a separation 'ν_30 large ⇒ refutation long' is exactly the bounded-arithmetic style of structural lower bound that escapes both NATURAL_PROOFS (ν_30 depends on G, not on a truth-table) and ALGEBRIZATION (no ring extension respects vertex-separator counts).

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_41e6d60a.py", line 194, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_41e6d60a.py", line 155, in run_trial
    nu_30 = compute_nu_30(edges, n)
            ^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_41e6d60a.py", line 68, in compute_nu_30
    nu_values.append(compute_min_vertex_separator(subgraph_edges, k))
                     ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_41e6d60a.py", line 46, in compute_min_v
    if not is_connected(edges, n):
           ^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_41e6d60a.py", line 35, in is_connected
    graph[v].add(u)
    ~~~~~^^^
KeyError: 7
```

Judge reasoning

The test crashed with a KeyError in is_connected before producing any RESULT line, so neither support nor falsification conditions can be evaluated. The critic also challenged due to lack of data. | next: Fix the graph construction in is_connected (initialize adjacency entries for all vertices in the induced subgraph, e.g., using defaultdict(set) or pre-populating keys) and rerun the pre-registered protocol.

Cubic-Form Lie-Stabilizer Codim Lower-Bounds DISJ Communication

- **Verdict**: `BARRIER_HIT`
- **Bridge**: Geometric Complexity Theory: Mulmuley–Sohoni Lie-algebra stabilizers of explicit cubic forms c_M ∈ $\mathrm{Sym}^{3(R}$N), accessed through the codimension κ(M)=N^2 − dim Stab_{gl_N}(c_M). This is a first-order/infinitesimal GCT invariant distinct from (a) Kronecker-coefficient counts and (b) full orbit-closure dimensions, the two GCT-COMM bridges that previously hit ALGEBRIZATION in this approach; <5 papers connect cubic-form stabilizers to communication complexity. × Randomized two-party communication complexity CC_R(M) of Boolean matrices M ∈ {0,1}^{N×N}, with the target M = M_DISJ_n on N = 2^n inputs (Razborov 1992 Ω(n) bound), instantiating the COMM_DISJ tensor-invariant program.
- **Recorded**: 2026-05-17 10:38 UTC
- **Entry ID**: `9d129ecb4a7f`

Statement

For $M \in \{0,1\}^{N\times N}$ ($N = 2^n$), form the symmetric cubic form $c\_M(x) := \Sigma\_\{i,j,k\in[N]\} M[i,j]\cdot M[j,k]\cdot M[i,k]\cdot x\_i\ x\_j\ x\_k$ whose coefficient at the symmetrized monomial $x_i\ x_j\ x_k$ is the count of directed triangles (i,j,k) in the bipartite-merged graph of M. Let $g(M) := \dim\{A \in gl\_N :$ the polynomial $\Sigma\_i\ (Ax)\_i \cdot \partial c\_M/\partial x\_i$ is identically zero$\}$, and define the GCT stabilizer codimension $\kappa(M) := N^2 - g(M)$. Conjecture: (a) for every Boolean M, $CC\_R(M) \geq (1/8)\cdot \log\_2(1 + \kappa(M))$, refuted by any single matrix with a randomized log-rank-style upper bound $CC\_R(M) \leq 2\cdot \log\_2(rank\_R(M)) + 1$ violating $(1/8)\cdot \log\_2(1+\kappa(M))$ by more than 1; (b) $\kappa(M\_DISJ\_n) \geq N - N^{3/4}$ for all $n \geq 3$, so (a) certifies $CC\_R(DISJ\_n) \geq n/8 - O(1) = \Omega(n)$.

Rationale

The triangle-count cubic form c_M encodes a non-polynomial-extension-friendly combinatorial feature of M (counting 3-cycles in its associated digraph) that is invisible to algebraic oracles, addressing why prior Kronecker/orbit-dimension GCT-COMM attempts hit ALGEBRIZATION. The infinitesimal Lie stabilizer is a first-order Mulmuley–Sohoni invariant that, unlike rank, is sensitive to triangle-richness — and DISJ's communication matrix has a sparse 3-disjoint structure ($|\{(S,T,U)$: pairwise disjoint$\}| = 3^n$ □ N^3) that should generically force a trivial stabilizer. The 1/8 constant tracks a sym-cubic Hilbert-series degeneracy factor and is conservative.

Novelty

- Judge: " over 0 arXiv hits

Empirical Test

- exit code: `-1`, elapsed: `0.00s`

```
(no output)
```

Judge reasoning

Rejected by ALGEBRIZATION barrier

Discrete Morse Critical Cells of Conflict Complex Bound DPLL Tree Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Discrete Morse theory on simplicial complexes (Forman 1998; Joswig–Pfetsch 2005 greedy lex matching; Benedetti–Lutz 2014 random discrete Morse) — a corner of combinatorial topology essentially absent from proof complexity (arXiv 'discrete Morse' AND ('Resolution' OR 'DPLL' OR 'proof complexity') returns no direct hits, with <5 adjacent independence-complex papers) × DPLL / tree-like Resolution refutation tree size t*(F) for unsatisfiable 3-CNFs, in the Cook–Reckhow–Krajíček bounded-arithmetic regime where ¬F is a Σ^b_0 statement and log_2 t*(F) controls V^0-witnessing complexity of unsatisfiability
- **Recorded**: 2026-05-17 11:47 UTC
- **Entry ID**: `02fea1709cc2`

Statement

For an unsat 3-CNF F with clauses $c_1,\ldots,c_m$, let C_F be the clause-conflict graph (vertex per clause; edge $\{c_i,c_j\}$ iff they share at least one complementary literal pair) and let $I\_\{\leq 3\}(C\_F)$ be the independence complex of C_F truncated to faces of size ≤ 4 (faces = clause-subsets pairwise free of complementary-literal conflicts). Run Joswig–Pfetsch greedy lex discrete-Morse matching: traverse simplices in (dim, lex) order, pair each σ with its lex-smallest currently-unmatched coface τ of dim(σ)+1 inside the truncation (otherwise mark σ critical). Let δ(F) := number of critical cells of dimension ≥1. Conjecture: there is an absolute constant c ≥ 1/4 such that for every unsat 3-CNF F with m ≥ 4, log_2(t*(F)) ≥ c · log_2(δ(F)+2), where t*(F) is the DPLL refutation tree-leaf count under lex-variable branching with unit propagation. A single F with measured log_2(t*(F)) < 0.25·log_2(δ(F)+2) refutes it.

Rationale

Critical cells of I(C_F) count obstructions to extending a locally-consistent partial certificate (a set of mutually non-resolving clauses) to a global one, which is precisely the local-to-global obstruction Ben-Sasson–Wigderson width-size trade-offs quantify for DPLL. A logarithmic relation is the natural scale: t* grows exponentially in hardness while δ grows exponentially in conflict-topological richness, so log–log alignment is plausible. The bridge avoids algebrization (Forman matching is purely combinatorial — no ring, no polynomial extension), evades natural proofs (the invariant is syntactic on F, not on its truth table), and lies in a regime Razborov/Krajíček bounded-arithmetic witnessing leaves open.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `0.04s`

```
7}
TRIAL: {'metric_name': 'log2(t*)/log2(δ+2)', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture
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TRIAL: {'metric_name': 'log2(t*)/log2(δ+2)', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'log2(t*)/log2(δ+2)', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture
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TRIAL: {'metric_name': 'log2(t*)/log2(δ+2)', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'log2(t*)/log2(δ+2)', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'log2(t*)/log2(δ+2)', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'log2(t*)/log2(δ+2)', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'log2(t*)/log2(δ+2)', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'log2(t*)/log2(δ+2)', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture
TRIAL: {'metric_name': 'log2(t*)/log2(δ+2)', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture
RESULT: INCONCLUSIVE reason=no_valid_instances
```

Judge reasoning

parse_fail | next: NONE

Gate-Cone Poset Mobius Value Lower-Bounds ACC0 Size for MOD-q

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Mobius functions on incidence algebras of finite posets (Rota 1964 combinatorial inversion; Stanley EC1 Ch.3) applied to the gate-dependency poset of an ACC^0 circuit; an algebraic-combinatorial invariant essentially absent from circuit complexity (arXiv search ‘Mobius function poset’ AND ‘ACC^0’ returns no hits; <5 adjacent papers on poset invariants for circuits) × ACC^0[m] circuit size for explicit symmetric functions (notably MOD_q for q coprime to m) in the Williams 2011 / Murray-Williams 2018 algorithmic-method regime, where structural circuit invariants underlie speedup-style lower bounds (NEXP ⊄ ACC^0)

- **Recorded**: 2026-05-17 15:23 UTC
- **Entry ID**: `3b280d479739`

Statement

Let C be an ACC^0[m] circuit on n inputs with s gates and depth d. Adjoin a formal minimum 0̂ below all gates and let P(C) be the poset on {0̂} ∪ Gates(C) with g ≤ h iff h = g or there is a directed wire path g → h (and 0̂ < g for every gate g). Define M(C) := |μ_{P(C)}(0̂, g_out)| where μ is the Mobius function of P(C) and g_out is the output gate, computed by μ(0̂,0̂)=1 and μ(0̂,h) = −Σ_{0̂ ≤ z < h} μ(0̂,z). Conjecture (A, upper anchor): M(C) ≤ (s+1)^d for every ACC^0[m] circuit. Conjecture (B, hardness anchor): for every ACC^0[m] circuit C that correctly computes MOD_q on n inputs with gcd(q,m)=1, one has M(C) ≥ $2^{\{c \cdot n}$\{1/(d+1)}} for an absolute constant c ≥ 1/8; a single counterexample circuit refutes (B).

Rationale

Mobius inversion on the gate-dependency poset is a CIRCUIT invariant (not a truth-table property), so it dodges the natural-proofs trap that killed the Plunnecke variant in this approach; it is also non-ring (no polynomial extension makes the recursion μ(0̂,h)=−Σμ(0̂,z) meaningful over F_p[X]), so it should not algebrize. Heuristically, Smolensky's low-degree-polynomial obstruction for MOD_q in ACC^0[m] forces long, non-factorisable dependency chains in any small circuit, which inflate alternating-sign Mobius cancellations on antichains of the gate poset; the M(C) invariant captures this 'inversion depth' more sharply than gate count alone and is computable in O(s^2) time, making it a viable Williams-style structural witness.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_fbcfc841.py", line 123, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_fbcfc841.py", line 78, in run_trial
    circuit = generate_random_circuit(n, s, d, seed)
              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_fbcfc841.py", line 27, in generate_rand
    num_inputs = random.randint(2, min(3, len(gates)))
                 ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/lib/python3.12/random.py", line 336, in randint
    return self.randrange(a, b+1)
           ^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/lib/python3.12/random.py", line 319, in randrange
    raise ValueError(f"empty range in randrange({start}, {stop})")
ValueError: empty range in randrange(2, 1)
```

Judge reasoning

The test crashed with a ValueError in generate_random_circuit (empty range in randrange(2,1)) before any instances were evaluated, so neither the support nor falsification condition could be assessed. | next: Fix the circuit generator to guarantee len(gates) ≥ 2 (or clamp the fan-in lower bound) before sampling num_inputs, then rerun the pre-registered sweep.

F_2-Corank of Minterm Incidence Bounds Monotone Formula Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Matroid theory — F_2-linear matroid corank (dimension of left F_2-null space) of the minterm-variable incidence matrix of a monotone Boolean function, viewed as a functional combinatorial-matroid invariant in the Oxley / Brualdi tradition of binary-matroid representability. This bridge to monotone formula complexity has essentially no presence in the literature (arXiv 'minterm incidence F_2 corank' AND 'monotone formula' returns no direct hits, <5 adjacent Razborov-approximator papers). × Monotone formula leaf-size L(f) for the k-CLIQUE indicator on K_v (n = v(v-1)/2 edge variables) in the Razborov 1985 regime, with k = 3 fixed and the n$^{\mathrm{Omega(k}}${1/4}) monotone lower bound as the prototypical target for the MONOTONE_CLIQUE focus.
- **Recorded**: 2026-05-17 15:55 UTC
- **Entry ID**: `dfc077cb4c36`

Statement

For a monotone Boolean function f on n variables, let M(f) ⊂ {0,1}^n be its set of minterms (each minterm identified with the 0/1 incidence vector of its support set), let A(f) ∈ F_2^{|M(f)| × n} be the minterm-variable incidence matrix, and define μ_2(f) := |M(f)| − rank_{F_2}(A(f)) (the F_2-matroid corank of the row matrix; equivalently the dimension of the left F_2-null space). We conjecture (a) μ_2(f) ≤ L(f)·log_2(L(f)+2) for every monotone Boolean function f, where L(f) is the minimum monotone formula leaf-size of f; and (b) for the k=3 CLIQUE indicator on K_v, μ_2(f) = C(v,3) − (v−1)(v−2)/2 = (v−1)(v−2)(v−3)/6 = Ω(v^3) = Ω(n^{3/2}). Combined, (a)+(b) imply L(CLIQUE_{v,3}) = Ω(v^3 / log v); a single monotone f with μ_2(f) > L(f)·log_2(L(f)+2) refutes (a), and any v ∈ {4,…,8} with μ_2(CLIQUE_{v,3}) ≠ (v−1)(v−2)(v−3)/6 refutes (b).

Rationale

F_2-corank of the minterm-variable incidence is a purely combinatorial matroid invariant that quantifies forced linear redundancies in the minterm cloud — exactly the kind of Steiner-design structure that the triangle hypergraph of K_v exhibits because each edge sits in v−2 triangles. Razborov's approximator philosophy already hints that bottom-up monotone formula construction cannot manufacture such global multiplicative redundancy without paying in leaves; turning that intuition into a clean matroid-rank bound (a) would yield a poly-time-computable, monotone-formula-tight, Steiner-aware lower-bound measure that contributes directly to the MONOTONE_CLIQUE target while staying inside the algebraic-rank toolkit that previously yielded the SUPPORTED 'Tropical Rank of Clause-Indicator' result.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ca306781.py", line 100, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ca306781.py", line 88, in run_trial
    "metric_value": sum(r["mu_2"] for r in results),
                    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ca306781.py", line 88, in <genexpr>
    "metric_value": sum(r["mu_2"] for r in results),
                        ~^^^^^^^^
KeyError: 'mu_2'
```

Judge reasoning

The test crashed with a KeyError: 'mu_2' before producing any data, so no RESULT line was emitted and neither the support nor falsification condition can be evaluated. | next: Fix the test harness so each result dict includes the 'mu_2' key (or guard the sum with .get), then re-run the pre-registered protocol over v∈{4..8} and the random monotone/TH_k^n configs.

Hypercontractive Laplacian-Spectrum Flatness Bounds Spectral SOS Max-Cut Ratio

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Fourier-analytic hypercontractive flatness invariants of graph Laplacian spectra (4-norm-to-2-norm ratios in the Bonami–Beckner–Bourgain tradition, applied as a 'pseudo-Fourier-level-2' invariant of the degree-2 SOS moment-matrix proxy) × Spectral / degree-2 SOS moment-matrix relaxation of Max-Cut on random 3-regular graphs (Delorme–Poljak spectral upper bound (n/4)λ_max as the canonical degree-2 SOS surrogate; integrality gap target ≥ 0.121)
- **Recorded**: 2026-05-17 16:22 UTC
- **Entry ID**: `c8a7a640d375`

Statement

For a 3-regular graph G on n vertices with Laplacian L = D − A having eigenvalues 0 = λ_n < λ_{n−1} ≤ … ≤ λ_1, define the hypercontractive Laplacian flatness HCF(G) := (n−1) · (Σ_{k=1}^{n−1} λ_k^4) / (Σ_{k=1}^{n−1} λ_k$^{2)}$2 ∈ [1, n−1], and the spectral degree-2 moment-matrix Max-Cut ratio ρ(G) := MC(G) / SB(G), where SB(G) := (n/4)·λ_1 and MC(G) is the true Max-Cut. Conjecture: for every 3-regular graph G with n ≥ 12, ρ(G) ≤ 1 − (1/10)·((HCF(G) − 1)/(n − 1))^{1/2}. A single 3-regular graph violating this strict inequality refutes the conjecture; in particular, any spectral-SOS 0.879-approximator would require HCF(G) < 1 + (10·0.121)^2·(n−1).

Rationale

The Laplacian spectrum is exactly the eigenvalue spectrum of the rank-1 degree-2 SOS moment-matrix surrogate B ∝ v_1 v_1^T scaled by λ_1, and its 4-vs-2-norm ratio is the natural Boolean hypercontractive flatness invariant; a spectrum concentrated at the top (HCF ≈ 1) corresponds to a 'junta-like' moment matrix where the spectral relaxation is tight, while a spread spectrum (HCF 1) forces an integrality gap exactly because the dominant eigenvector loses bipartite alignment with G's edges. Restricting to 3-regular graphs avoids the dense-graph (K_n) and exact-cycle pathologies where the spectral bound is tight identically. The bridge between hypercontractive flatness of Laplacian spectra and Max-Cut integrality gaps appears to have <5 papers (Friedgut/Bourgain-style hypercontractivity has not been formally aimed at SOS Max-Cut moment matrices).

Novelty

- Judge: `NOVEL` over 4 arXiv hits

Top hits consulted: - [1107.1630v3] On the Integrality Gap of the Subtour LP for the 1,2-TSP - [1207.5696v1] Beyond Max-Cut: λ-Extendible Properties Parameterized Above the Poljak-Turzík Bound - [2403.19550v2] Spectral gap for surfaces of infinite volume with negative curvature - [s2:1911.01411] Lifting sum-of-squares lower bounds: degree-2 to degree-4

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_67b46128.py", line 194, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_67b46128.py", line 148, in run_trial
    eigenvalues = sorted(matrix_eigenvalues(L))
                         ^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_67b46128.py", line 55, in matrix_eigenv
    Q[i][j] = Fraction(v[i], norm)
              ^^^^^^^^^^^^^^^^^^^^
  File "/usr/lib/python3.12/fractions.py", line 277, in __new__
    raise TypeError("both arguments should be "
TypeError: both arguments should be Rational instances
```

Judge reasoning

The test crashed with a TypeError in matrix_eigenvalues before any of the 150 instances were evaluated, so no data exists to assess support_fraction or min_margin. Per rule 2, a crash yields INCONCLUSIVE. | next: Fix matrix_eigenvalues to use floating-point (numpy.linalg.eigvalsh) or coerce the norm to a Rational/sqrt-friendly type before re-running the 150-instance sweep.

Free Cumulant Excess of Laplacian Spectrum Bounds Max-Cut SoS-2 Gap

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Free probability of Voiculescu-Speicher — non-crossing partition cumulants of the empirical spectral measure of a signed graph Laplacian, computed by Möbius inversion on the lattice NC(k). This corner of operator-algebraic combinatorics has essentially no presence in CSP / SoS integrality-gap literature (an arXiv search for 'free cumulant' AND ('Max-Cut' OR 'integrality gap' OR 'Sum-of-Squares') returns no direct hits and <5 adjacent free-probability-meets-graph-spectra papers, all on Wigner-type ensembles). × Degree-2 Sum-of-Squares / Lasserre integrality gap for Max-Cut on random 3-regular graphs (the Goemans–Williamson / Delorme–Poljak regime, target of Schoenebeck-Khot-Vishnoi-Tulsiani SoS integrality-gap programs).
- **Recorded**: 2026-05-17 16:51 UTC
- **Entry ID**: `c10c070e26bd`

Statement

For a 3-regular graph G on $n \geq 8$ vertices, let $\hat{L}_G = (D-A)/3$ be the normalized Laplacian, μ_G its empirical spectral measure, and let $\kappa_j(G)$ denote its jth free cumulant computed by Möbius inversion on NC(j) ($\kappa_2 = m_2 - m_1^2$, $\kappa_4 = m_4 - 4 m_3 m_1 - 2 m_2^2 + 10 m_2 m_1^2 - 5 m_1^4$). Define the free-Poisson excess $\varepsilon(G) := \max(0, \kappa_4(G)/\kappa_2(G)^2 - 1)$ and the gap $g(G) := SDP_2(G)/MaxCut(G)$, where $SDP_2(G) = (3n/8)\cdot(1 + \lambda_{max}(\hat{L}_G - I))$ is the Delorme–Poljak spectral SoS-2 upper bound. CONJECTURE: for every 3-regular G, $(g(G) - 1) \cdot \log_2(n) \leq 4 \cdot (\varepsilon(G) + 1/n)$. A single instance violating this bound, or any instance with $g(G) > 1 + 4\cdot(\varepsilon(G)+1/n)/\log_2 n$, falsifies the conjecture.

Rationale

Free cumulants linearize free convolution and detect deviation from a Marchenko–Pastur / free-Poisson reference, so $\varepsilon(G)$ measures exactly the bulk-spectrum 'non-randomness' that drives

Delorme–Poljak slack on sparse random Cayley-like graphs. Because SDP_2 is a strict function of the Laplacian spectrum while MaxCut depends on combinatorial cycle structure, any controlled bound on (SDP_2/MaxCut − 1) by a spectral free-probability quantity offers a CSP-instance witness rather than a truth-table predicate, avoiding the NATURAL_PROOFS trap that killed the analogous Lasserre moment-rank attempt. The bridge is novel: free cumulants of graph Laplacians have been studied in random-matrix limits but never used as a finite-n SoS-gap certifier.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `5.32s`

```
conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'T(G)', 'metric_value': -2.125814583693911, 'instances_tested': 1, 'conjectur
TRIAL: {'metric_name': 'T(G)', 'metric_value': -2.125814583693911, 'instances_tested': 1, 'conjectur
TRIAL: {'metric_name': 'T(G)', 'metric_value': -2.25, 'instances_tested': 1, 'conjecture_holds': Tru
TRIAL: {'metric_name': 'T(G)', 'metric_value': -2.060964047443681, 'instances_tested': 1, 'conjectur
TRIAL: {'metric_name': 'T(G)', 'metric_value': -2.060964047443681, 'instances_tested': 1, 'conjectur
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TRIAL: {'metric_name': 'T(G)', 'metric_value': -2.125814583693911, 'instances_tested': 1, 'conjectur
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TRIAL: {'metric_name': 'T(G)', 'metric_value': -2.25, 'instances_tested': 1, 'conjecture_holds': Tru
TRIAL: {'metric_name': 'T(G)', 'metric_value': -2.060964047443681, 'instances_tested': 1, 'conjectur
TRIAL: {'metric_name': 'T(G)', 'metric_value': -2.25, 'instances_tested': 1, 'conjecture_holds': Tru
TRIAL: {'metric_name': 'T(G)', 'metric_value': -2.060964047443681, 'instances_tested': 1, 'conjectur
TRIAL: {'metric_name': 'T(G)', 'metric_value': -2.060964047443681, 'instances_tested': 1, 'conjectur
RESULT: SUPPORTED mean=-2.1187349498381076 std=0.08924677387309159 support_fraction=1.0
```

Judge reasoning

Although the test's RESULT line reports SUPPORTED with T(G)≤0 on all 150 instances, the critic's challenge is well-founded: T values cluster near −2.1, indicating g(G)≈1 on these small near-bipartite 3-regular graphs, so the bound is satisfied trivially rather than meaningfully tested. The pre-registered regime (n≤16) does not probe the adversarial large-n regime where the SDP-MaxCut gap actually opens. | next: Re-run on larger 3-regular graphs (n ∈ {50, 100, 200, 500}) and include known integra

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Schatten-1/Schatten-4 Ratio Lower-Bounds AND-Function Communication

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Noncommutative L^p geometry — Schatten-p norm ratios in the Pisier / Lust-Piquard framework, specifically the (S_1, S_4) stable-rank invariant τ(M) = log_2(||M||*{S_1}^2 / ||M*||{S_4}^2). This corner of operator-space theory is rarely applied to communication complexity beyond Forster's operator-norm sign-rank bound; an arXiv search for 'Schatten 4 norm' AND 'communication complexity' returns <5 hits, all on unrelated invariants (e.g., spectral norm or γ_2). × Randomized two-party communication complexity of AND-functions M_g(x∧y) on {0,1}^n × {0,1}^n, with the prototype g = AND-of-NOTs giving M = M_DISJ_n (Razborov 1992 Ω(n) regime). The conjecture instantiates the COMM_DISJ tensor-invariant program and parallels the spectral-expansion mechanism underlying Tseitin-on-expander resolution width lower bounds (Ben-Sasson–Wigderson, Atserias–Bonet–Esteban).
- **Recorded**: 2026-05-17 17:24 UTC

- **Entry ID**: `87f5d4a9eef6`

Statement

For g: {0,1}^n → {0,1}, let M_g ∈ {0,1}$^{\{2}$n × 2^n} be the AND-function communication matrix M_g[x,y] = g(x ∧ y). With σ_i = singular values of M_g, define τ(M_g) := log_2((Σ_i σ_i)^2 / (Σ_i σ_i$^{4)}${1/2}) = 2·log_2(||M_g||*{S_1}) − log_2(||M_g*||{S_4}^2). Conjecture (A): for every AND-function, CC_R^{1/3}(M_g) ≥ (1/8) τ(M_g) − 2; (B): for M_DISJ_n (g(z) = ∏_i(1−z_i)), τ(M_DISJ_n) = n·log_2(5/√7) ≈ 0.918n, exactly. A single AND-function M_g with τ(M_g) > 8·D^cc(M_g) + 16, or any n with τ(M_DISJ_n) < 0.7n, falsifies.

Rationale

The DISJ matrix factors as a tensor power of the 2×2 block (1,1;1,0) with golden-ratio spectrum {φ, −1/φ}, so its Schatten norms have closed forms (||·||*{S_p}^p = (φ^p + φ*$^{\{-p\})}$*n), giving τ = Θ(n) by exact computation rather than spectral inequalities. The (S_1, S_4) ratio strengthens Forster's (S_2, S*∞) ratio: it is more sensitive to mid-spectrum mass while remaining sub-multiplicative under tensor product, the same algebraic property that makes Tseitin-on-expanders amenable to width-via-expansion arguments. If proved, conjecture (A) yields a clean, calculable tensor invariant that matches Razborov's Ω(n) bound on a wide AND-function class and would import noncommutative L^p techniques into proof-complexity-flavored CC lower bounds.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.20s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_624caab1.py", line 177, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_624caab1.py", line 145, in run_trial
    tau = compute_tau(matrix)
          ^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_624caab1.py", line 84, in compute_tau
    u, s, vt = svd(matrix)
               ^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_624caab1.py", line 55, in svd
    s[i] = matrix_norm([row[i] for row in b])
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_624caab1.py", line 33, in matrix_norm
    return math.sqrt(sum(sum(x**2 for x in row) for row in a))
                     ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_624caab1.py", line 33, in <genexpr>
    return math.sqrt(sum(sum(x**2 for x in row) for row in a))
                            ^^^^^^^^^^^^^^^^^^^
TypeError: 'float' object is not iterable
```

Judge reasoning

The test crashed with a TypeError in the custom SVD/matrix_norm implementation before any data was produced, so neither the support nor falsification criteria can be evaluated. | next: Fix the matrix_norm/SVD helper (use numpy.linalg.svd on the 2D matrix instead of the buggy nested-iteration norm) and rerun the n=4..8 sweep.

Quiver Mutation Neighborhood Lower-Bounds ACC0 by Sensitivity

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Cluster algebra theory of Fomin–Zelevinsky — finite-step quiver mutation neighborhoods (FZ 2002 'Cluster algebras I'; Keller's mutation-class machinery). Mutations are non-ring combinatorial graph rewrites (arrow reversal at a vertex, path completion, 2-cycle cancellation) on directed multi-graphs; an arXiv search 'quiver mutation' AND ('ACC^0' OR 'circuit complexity' OR 'Boolean circuit') returns no direct hits, with <5 adjacent papers using cluster algebras for combinatorial gadgets — never for circuit lower bounds. × ACC^0[m] circuit size and depth for Boolean functions, in the Williams 2011 / Murray–Williams 2018 algorithmic-method regime targeting NEXP ⊄ ACC^0 and MOD_q lower bounds for q coprime to m. The invariant is a STRUCTURAL property of the circuit DAG (not of its truth table), aligning with Williams-style structural exploitations of small-circuit structure.
- **Recorded**: 2026-05-17 17:51 UTC
- **Entry ID**: `c7f898dc68a4`

Statement

For an ACC^0[m] circuit C with N gates and depth d computing f:{0,1}^n→{0,1} with average sensitivity I(f)=E_x[#{i:f(x⊕e_i)≠f(x)}], let Q(C) be the multi-quiver whose vertices are circuit nodes (inputs and gates) and whose arc multiplicity i→g equals the number of times node i feeds gate g, with 2-cycles cancelled. Define NbhdSize(C):=#{isomorphism classes of μ_v(Q(C)) : v∈V(Q(C))}, where μ_v is the Fomin–Zelevinsky mutation at v (reverse all arrows at v; for each path i→v→j add an arrow i→j with multiplicity mult(i→v)·mult(v→j); cancel resulting 2-cycles), and isomorphism is detected by Weisfeiler–Lehman color refinement on the labeled multi-digraph. Conjecture: NbhdSize(C) ≥ ⌈I(f)/(4(d+1))⌉ for every ACC^0[m] circuit; a single C with NbhdSize(C) < ⌈I(f)/(4(d+1))⌉ refutes the conjecture.

Rationale

Quiver mutation is a non-commutative, non-ring graph rewrite that cancels 2-cycles — an operation that does not lift coherently to polynomial extensions of the Boolean ring, placing the invariant outside the Aaronson–Wigderson algebrization frame that has tripped past GCT-flavoured ACC^0 attempts. NbhdSize is a structural circuit invariant rather than a truth-table predicate, so the Razborov–Rudich largeness/usefulness clauses do not apply and the natural-proofs barrier is sidestepped (the same f admits circuits with very different NbhdSize). Heuristically, high-sensitivity targets force the gate-multi-quiver to be mutation-rigid (many non-isomorphic 1-step neighbours), because low-rigidity quivers correspond to symmetric, low-sensitivity computation — giving a Williams-style structural-versus-functional tension.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6aabdc6d.py", line 160, in <module>
    result = run_trial(seed)
```

```
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6aabdc6d.py", line 136, in run_trial
    mutated_quiver = mutate_quiver(quiver, vertex)
                     ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6aabdc6d.py", line 57, in mutate_quiver
    for u in quiver:
RuntimeError: dictionary changed size during iteration
```

Judge reasoning

The test crashed with a RuntimeError (dictionary changed size during iteration) in mutate_quiver before any trials produced data, so neither the support nor falsification condition can be evaluated. | next: Fix mutate_quiver to iterate over a snapshot (e.g., list(quiver.items())) or build the mutated quiver in a fresh dict, then rerun the 90-trial pre-registered protocol.

Burau Trace Defect of Clause Braid Lower-Bounds Tree-Resolution Steps

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Braid-group representation theory — the unreduced Burau representation ρ_t: B_n → GL_n(Z[t,t^{-1}]) specialised at t = −1, where Burau factors through the integer symplectic group Sp_{2g}(Z) (Burau 1936; Squier 1984; Birman 1974), and the trace defect ω(β) := |n − tr(ρ_{-1}(β))|. This corner of knot/braid theory has essentially no presence in proof complexity (arXiv search 'Burau representation' AND ('Frege' OR 'resolution refutation' OR 'proof complexity') returns no direct hits; <5 adjacent papers). × Tree-like Resolution / DPLL refutation step count for unsatisfiable 3-CNFs, viewed as a Frege-depth proxy in the Cook–Reckhow–Krajíček bounded-arithmetic framework where ¬F is Σ^b_0 and log_2 of the canonical tree size controls V^0-witnessing complexity (Beame–Pitassi 2001; Ben-Sasson–Wigderson 2001).
- **Recorded**: 2026-05-17 18:27 UTC
- **Entry ID**: `b455093f5476`

Statement

For an unsat 3-CNF F on n variables with m clauses, construct a braid word β(F) ∈ B_n by reading clauses C ∈ F in lex order: for each C with literals l_1, l_2, l_3 sorted by |var|, append generators σ_{|l_1|}^{sgn(l_1)·sgn(l_2)} · σ_{|l_2|}^{sgn(l_2)·sgn(l_3)} (σ_k for exponent +1, σ_k^{-1} for exponent −1, with σ_k acting between strands k and k+1, k ≤ n−1; pairs that straddle strand n−1 are skipped). Let M(F) = ρ_{-1}(β(F)) be the unreduced Burau matrix product specialised at t = −1 (so each generator is an integer n × n matrix with entries in {−1, 0, 1, 2}), and ω(F) := |n − tr(M(F))|. Conjecture: there is an absolute constant C ≥ 1/4 such that the canonical lex-DPLL refutation backtrack count t(F) satisfies 4·(t(F)^2 + 1) ≥ ω(F) for every unsat 3-CNF F. A single unsat F with 4·(t(F)^2 + 1) < ω(F) falsifies the conjecture.

Rationale

At t = −1 the Burau representation lands inside Sp_{2g}(Z), so the matrix product realises clause-pair conflicts as integer symplectic transvections and the trace defect quantifies how entangled the resulting symplectic flow becomes — a relational, non-commutative invariant of the SYNTACTIC clause ordering that is invisible to width or expansion alone. Any tree-Resolution refutation must split on every variable participating in an irreducible conflict, so a highly entangled clause braid should force backtracking. The invariant is structural (CNF syntax, not truth-table), sidestepping NATURAL_PROOFS; braids are non-abelian and do not lift to polynomial extensions of the Boolean ring, sidestepping ALGEBRIZATION; the construction uses no diagonalisation, sidestepping RELATIVIZATION.

Novelty

- Judge: NOVEL over 2 arXiv hits

Top hits consulted: - [s2:68cab25640a4debd147cd412a08597d352b33] Refutation of a Proof Complexity Approach to Maxsat Resolution-Based Systems - [3506702] On Proof Complexity of Resolution over Polynomial Calculus

Empirical Test

- exit code: 0, elapsed: 0.04s


```
alue": 0, "instances_tested": 0, "conjecture_holds": false, "counterexample": "No valid instances gen
TRIAL: {"seed": 463, "metric_name": "omega", "metric_value": 0, "instances_tested": 0, "conjecture_ho
TRIAL: {"seed": 503, "metric_name": "omega", "metric_value": 0, "instances_tested": 0, "conjecture_ho
TRIAL: {"seed": 547, "metric_name": "omega", "metric_value": 0, "instances_tested": 0, "conjecture_ho
TRIAL: {"seed": 593, "metric_name": "omega", "metric_value": 0, "instances_tested": 0, "conjecture_ho
TRIAL: {"seed": 631, "metric_name": "omega", "metric_value": 0, "instances_tested": 0, "conjecture_ho
TRIAL: {"seed": 677, "metric_name": "omega", "metric_value": 0, "instances_tested": 0, "conjecture_ho
TRIAL: {"seed": 727, "metric_name": "omega", "metric_value": 0, "instances_tested": 0, "conjecture_ho
TRIAL: {"seed": 773, "metric_name": "omega", "metric_value": 0, "instances_tested": 0, "conjecture_ho
TRIAL: {"seed": 821, "metric_name": "omega", "metric_value": 0, "instances_tested": 0, "conjecture_ho
TRIAL: {"seed": 877, "metric_name": "omega", "metric_value": 0, "instances_tested": 0, "conjecture_ho
TRIAL: {"seed": 929, "metric_name": "omega", "metric_value": 0, "instances_tested": 0, "conjecture_ho
RESULT: FALSIFIED counterexample="No valid instances generated" first_failing_seed=11
```


Judge reasoning

All 30 seeds produced zero accepted instances ('No valid instances generated'), so the conjecture was never actually evaluated; the RESULT line's 'FALSIFIED' label reflects a generator failure rather than a genuine counterexample. Pre-registered acceptance threshold (n_accepted ≥ 400) was not remotely met. | next: Fix the instance generator (e.g., loosen acceptance, verify DPLL unsat check, ensure clauses produce non-skipped Burau generators at α=6, n∈{8..16}) and rerun to obtain a non-empty sam

---

Euler Characteristic of Gate-Conflict Independence Complex Bounds ACC^0[m] Size for MOD_q

- **Verdict**: INCONCLUSIVE
- **Bridge**: Topological combinatorics — reduced Euler characteristic $\tilde{\chi}$(Ind(G)) of the independence complex of a finite graph G, computed as the independence-polynomial value I(G;-1) (Kozlov 2008 'Combinatorial Algebraic Topology'; Engström 2009 'Complexes of directed trees and independence complexes'; Stanley-Reisner / Alexander duality). This corner of combinatorial algebraic topology has essentially no presence in circuit complexity (arXiv 'independence complex' AND ('ACC^0' OR 'circuit complexity' OR 'Williams algorithmic method') returns no direct hits; <5 adjacent monotone-circuit-matroid papers). × ACC^0[m] circuit size for MOD_q on n inputs with q coprime to m (focus q=3, m=2), in the Williams 2011 / Murray-Williams 2018 algorithmic-method regime targeting NEXP ⊄ ACC^0. The invariant is a STRUCTURAL property of the circuit DAG (gate-conflict graph), not of the truth table, so it is shielded from the Razborov-Rudich natural-proofs barrier.
- **Recorded**: 2026-05-17 19:02 UTC
- **Entry ID**: cdef279c90bb

Statement

For an ACC^0[m] circuit C of depth d and size s on n inputs, let G_C be the gate-conflict graph: vertices = non-input gates; (g,h) is an edge iff the input-variable supports supp(g), supp(h) ⊆ [n] satisfy supp(g)∩supp(h) ≠ ∅. Let τ(C) := |I(G_C;-1)| where I(G;x) = Σ_{S independent in G} x^{|S|}. Conjecture: every depth-d ACC^0[m] circuit C that ε-approximates MOD_q on n inputs (q coprime to m, ε ≤ 1/4, q≥3) satisfies τ(C) ≥ 2^{n/(10·d²)} − 1. A single (C, q, ε) instance with τ(C) < 2^{n/(10·d²)} − 1 yet agreement(C, MOD_q) ≥ (1−ε)·2^n refutes it.

Rationale

Razborov-Smolensky shows that approximating MOD_q in AC^0[p] forces large size via polynomial-method incompatibility; the independence complex of the gate-conflict graph categorifies the variable-overlap pattern that polynomial-degree bounds exploit, with χ̃ measuring 'how rigidly' supports must intersect to express many distinct minterms. Because τ(C) depends only on the wiring DAG (not the truth table), the invariant cannot be 'large + useful' in the Razborov-Rudich sense, dodging natural proofs; because it is built from set intersections and Möbius sign sums (not ring operations on tt-extensions), it does not algebrize in the Aaronson-Wigderson sense. The bound 2^{n/(10·d²)} aligns with Smolensky's √n degree barrier propagated through d layers.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [1904.05483v2] Parallels Between Phase Transitions and Circuit Complexity? - [1606.05050v1] Proof Complexity Lower Bounds from Algebraic Circuit Complexity - [9906008v2] A Lower Bound on the Average-Case Complexity of Shellsort - [2304.04810v2] Chain algebras of finite distributive lattices - [1110.6876v4] Lower bounds for topological complexity

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_cbad24de.py", line 135, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_cbad24de.py", line 103, in run_trial
    truth_table = compute_truth_table(circuit, n)
                  ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_cbad24de.py", line 45, in compute_truth
    output = input_values[gate_inputs[0]] ^ input_values[gate_inputs[1]]
                          ~~~~~~~~~~~^^^
IndexError: list index out of range
```

Judge reasoning

The test crashed with an IndexError before producing any data, so the pre-registered support condition could not be evaluated and no qualifying instances were assessed. | next: Fix the circuit-evaluation bug (guard gate_inputs indexing against gates with fewer than 2 inputs, or generate circuits whose gate fan-in matches the evaluator) and rerun the trial sweep.

Tamari Rotation Rank of Canonical DT Lower-Bounds XOR-Lifted F2-Rank

- **Verdict**: `INCONCLUSIVE`

- **Bridge**: Tamari lattice combinatorics — the order-theoretic Pallo rank function rrd(T) = sum over internal nodes v of |L(v)| (size of v's left-subtree leaf set), in the Sleator-Tarjan-Thurston / Pallo-Bonnin / Pournin / Chapoton tradition on Catalan-counted binary trees. An arXiv search for 'Tamari lattice' AND ('communication complexity' OR 'lifting theorem' OR 'decision tree complexity') returns no direct hits and <5 adjacent papers — this corner of Catalan/order combinatorics has essentially no presence in CC / lifting literature. × F2-rank of the XOR-lifted communication matrix M_{f xor} for Boolean f: {0,1}^n -> {0,1}, defined by M[a,b] = f(a XOR b), in the Raz-McKenzie / Goos-Pitassi-Watson lifting framework. log2 rank_{F2}(M) is the canonical algebraic lower bound for deterministic two-party CC^D(M_{f xor}), the XOR-function CC studied since Lee-Shraibman, Hatami-Hosseini-Lovett, and the Log-Rank Conjecture.
- **Recorded**: 2026-05-17 19:17 UTC
- **Entry ID**: `b045e08ff6c7`

Statement

For a Boolean function f: {0,1}^n -> {0,1}, let T_f* be the canonical minimum-leaf decision tree obtained by a deterministic DP on partial assignments: at each restriction state, branch on the smallest-index variable whose two children have strictly smaller leaf totals, with lex tie-breaking on serialized (left-shape, right-shape) tuples. Let rrd(T_f*) = sum over internal nodes v of |L(v)|, the Pallo right-rotation rank in the Tamari lattice. Define M = M_{f xor}, the 2^n by 2^n matrix M[a,b] = f(a XOR b). Conjecture: there is a universal constant c >= 1/4 such that for every f with rrd(T_f*) >= 4, log2(rank_{F2}(M)) >= c * log2(1 + rrd(T_f*)). A single f with rrd(T_f*) >= 4 but log2(rank_{F2}(M)) < 0.25 * log2(1 + rrd(T_f*)) refutes the conjecture.

Rationale

Raz-McKenzie / GPW lifting passes the STRUCTURE of a decision tree T_f into the rectangle/block structure of M_{f o g}; the Pallo rank rrd(T_f*) is the most natural Catalan-order measure of how 'right-comb-asymmetric' that block structure is, in a regime never previously connected to CC. The invariant is intrinsic to the chosen minimum tree T_f* (a non-polynomial selection over restrictions), so it is not a polynomial in truth-table entries and the construction does not extend to polynomial extensions of the Boolean ring — sidestepping algebrization. The bridge respects LIFTING's safe directions (no relativization, no Karp-Lipton) and targets a CC-rank object, not a P/poly truth-table predicate, so natural-proofs does not bite.

Novelty

- Judge: `NOVEL` over 2 arXiv hits

Top hits consulted: - [s00037-025-00276-5] Lifting Dichotomies - [s2:7250a91643b33e65c107da27b0c02900b7
Lifting Theorems for Equality

Empirical Test

- exit code: `0`, elapsed: `0.11s`


```
ces_tested": 1, "conjecture_holds": true, "counterexample": ""}
TRIAL: {"seed": 503, "metric_name": "log2(rank_F2(M)) / log2(1 + rrd(T_f*))", "metric_value": 0.79295
TRIAL: {"seed": 547, "metric_name": "log2(rank_F2(M)) / log2(1 + rrd(T_f*))", "metric_value": 0.73395
TRIAL: {"seed": 593, "metric_name": "log2(rank_F2(M)) / log2(1 + rrd(T_f*))", "metric_value": 0.99628
TRIAL: {"seed": 631, "metric_name": "log2(rank_F2(M)) / log2(1 + rrd(T_f*))", "metric_value": 0.83023
TRIAL: {"seed": 677, "metric_name": "log2(rank_F2(M)) / log2(1 + rrd(T_f*))", "metric_value": 0.73395
TRIAL: {"seed": 727, "metric_name": "log2(rank_F2(M)) / log2(1 + rrd(T_f*))", "metric_value": 0.99119
TRIAL: {"seed": 773, "metric_name": "log2(rank_F2(M)) / log2(1 + rrd(T_f*))", "metric_value": 0.99119
TRIAL: {"seed": 821, "metric_name": "log2(rank_F2(M)) / log2(1 + rrd(T_f*))", "metric_value": 0.85577
TRIAL: {"seed": 877, "metric_name": "log2(rank_F2(M)) / log2(1 + rrd(T_f*))", "metric_value": 0.79295
TRIAL: {"seed": 929, "metric_name": "log2(rank_F2(M)) / log2(1 + rrd(T_f*))", "metric_value": 0.79295
```

```
RESULT: SUPPORTED mean=0.8822973003045711 std=0.09969545211010329 support_fraction=1.0
```

Judge reasoning

Random-seed sampling reports mean ratio 0.88 with min 0.73, but the critic identifies a structured adversarial case (parity function) where rrd ~ n·2^n while F2-rank of the XOR-lift is 2, yielding ratio ≈0.166 < 0.25 for n=4 — not probed by the random sample. Per the strict rule, an unrefuted CHALLENGE downgrades the verdict. | next: Explicitly evaluate the conjecture on the parity function f(x)=x_1⊕...⊕x_n for n in {4,5,6,7} and other low-F2-rank structured functions (e.g., affine, inner product)

Immanant Log-Variance Separates Random Matrices from Det-Padded Blocks

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Symmetric-group representation theory: the Specht-character immanant vector imm_lambda(M)=sum_sigma chi^lambda(sigma) prod_i M[i,sigma(i)] as a partition-indexed structural decomposition of a matrix function (Stembridge 1991; Stanley EC2 Ch.7; Greene's immanant-positivity program). The invariant is RELATIONAL (vector-valued across p(n) partitions, not a single-number truth-table functional) and distinct from the blacklisted positivity-width attempt: the present invariant is the LOG-MAGNITUDE VARIANCE across the full Specht spectrum on PADDING-vs-RANDOM ensembles, an exp(n)-time meta-complexity-style discriminator, not a natural-proof-style poly-time invariant. × Meta-complexity in the GCT_DET_PERM regime: distinguishing K^t-random Boolean matrices from low-K^t / orbit-closure-of-det_m^{padded} matrices via a super-polynomial structural test, in the spirit of Hirahara's GapMCSP and the Mulmuley-Sohoni orbit-closure separation of perm_n from det_m^{poly}. The invariant V(M) is a candidate meta-complexity measure that lower-bounds compressibility under linear/block padding without being itself in P.
- **Recorded**: 2026-05-17 19:47 UTC
- **Entry ID**: `88defc39f508`

Statement

For M in {0,1}^{n x n} define I_lambda(M) := sum_{sigma in S_n} chi^lambda(sigma) prod_i M[i,sigma(i)] using the Murnaghan-Nakayama character table, and set V(M) := Var_{lambda in P(n)} (log_2(1 + |I_lambda(M)|)). Conjecture: for each n in {5,6,7} there exist absolute constants c1 >= 4*c2 > 0 such that (i) over uniform M in {0,1}^{n x n}, V(M) >= c1*(log_2 n)^2 on >= 27 of 30 seeds, while (ii) over the det-padded ensemble M_pad = block-diag(M', J_{n-m,n-m}) with m = ceil(n/2), M' uniform in {0,1}^{m x m} and J all-ones, V(M_pad) <= c2*(log_2 n)^2 on >= 27 of 30 seeds. A single (n, seed) pair where V(uniform) < 4*V(padded) on independent draws falsifies the conjecture.

Rationale

Padding a smaller block by an all-ones submatrix forces the immanant sum to factor through the permanent of the J-block, collapsing the Specht spectrum onto partitions adjacent to (n) and (1^n) and zeroing most middle-lambda components by row/column character cancellation. Uniform random Boolean matrices instead distribute i.i.d. signed contributions across all p(n) Specht coordinates, giving Theta(sqrt(n!)) magnitudes per lambda and high log-variance. The bridge from partition-indexed Specht variance to the Mulmuley-Sohoni padding question parallels how stabilizer-codimension invariants stratify det vs perm: padding by structureless blocks erases the representation-theoretic spread that random matrices exhibit.

Novelty

- Judge: `NOVEL` over 6 arXiv hits

Top hits consulted: - [1005.0979v1] Supersymmetry in Random Matrix Theory - [0911.5645v1] Non-Hermitian Random Matrix Ensembles - [0904.1432v2] Applications of random matrix theory to condensed matter and optical physics - [1611.00827v2] Geometric complexity theory and matrix powering - [1911.03990v1] Implementing geometric complexity theory: On the separation of orbit closures via symmetries

Empirical Test

- exit code: `0`, elapsed: `6.81s`

```
_name': 'log_variance_ratio', 'metric_value': 0.0, 'instances_tested': 3, 'conjecture_holds': True,
TRIAL: {'metric_name': 'log_variance_ratio', 'metric_value': 0.0, 'instances_tested': 3, 'conjecture
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TRIAL: {'metric_name': 'log_variance_ratio', 'metric_value': 0.0, 'instances_tested': 3, 'conjecture
TRIAL: {'metric_name': 'log_variance_ratio', 'metric_value': 0.0, 'instances_tested': 3, 'conjecture
TRIAL: {'metric_name': 'log_variance_ratio', 'metric_value': 0.0, 'instances_tested': 3, 'conjecture
TRIAL: {'metric_name': 'log_variance_ratio', 'metric_value': 0.0, 'instances_tested': 3, 'conjecture
TRIAL: {'metric_name': 'log_variance_ratio', 'metric_value': 0.0, 'instances_tested': 3, 'conjecture
TRIAL: {'metric_name': 'log_variance_ratio', 'metric_value': 0.0, 'instances_tested': 3, 'conjecture
TRIAL: {'metric_name': 'log_variance_ratio', 'metric_value': 0.0, 'instances_tested': 3, 'conjecture
TRIAL: {'metric_name': 'log_variance_ratio', 'metric_value': 0.0, 'instances_tested': 3, 'conjecture
TRIAL: {'metric_name': 'log_variance_ratio', 'metric_value': 0.0, 'instances_tested': 3, 'conjecture
TRIAL: {'metric_name': 'log_variance_ratio', 'metric_value': 0.0, 'instances_tested': 3, 'conjecture
RESULT: SUPPORTED mean=0.0000 std=0.0000 support_fraction=1.0000
```

Judge reasoning

Every trial reports metric_value=0.0 with std=0.0 and only 3 instances per seed rather than the pre-registered 30, so the two-sided inequality (V_uniform >= c1*(log n)^2 and V_padded <= c2*(log n)^2) is never actually evaluated. The degenerate identically-zero 'log_variance_ratio' indicates the test does not compute the conjectured quantity, so SUPPORTED cannot be trusted. | next: Reimplement the test to explicitly compute V(uniform) and V(padded) per seed via the Murnaghan-Nakayama character ta

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Hypercontractive Term-Support Stable Rank Bounds Monotone Clique DNF

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Fourier analysis / hypercontractivity — Bonami–Beckner noise-operator Gram kernels on DNF-term SUPPORT sets, evaluated through the stable rank $\sigma(K)=(\Sigma\lambda\_i)^2/\Sigma\lambda\_i^2$ of the resulting term-correlation matrix, in the Kahn–Kalai–Linial / Friedgut / Mossel tradition but used as a SYNTACTIC DNF invariant (a property of the formula, not its truth table) — an angle with essentially no presence in monotone-circuit lower-bound literature outside Razborov-approximator analogies × Monotone DNF / formula complexity for the k-CLIQUE indicator on v-vertex graphs (Razborov 1985 regime, target measure μ on monotone DNFs satisfying (i) submodular under conjunction, (ii) O(log n) for poly-size DNFs, (iii) Ω(n)/Ω(v) for the k-CLIQUE indicator), accessed through the canonical minterm DNF where each term is an edge-set of a k-clique
- **Recorded**: 2026-05-17 20:14 UTC
- **Entry ID**: `fad53ef695ab`

Statement

For a monotone DNF F = ∨*{i=1}^{s} T_i on N variables with each term T_i =* ∧{k∈S_i} x_k indexed by its support set S_i ⊆ [N], define the Bonami–Beckner term-correlation Gram matrix K(F) ∈ ℝ^{s×s} at noise parameter ρ=1/2 by K(F)[i,j] := 2^{-|S_i|-|S_j|} · (3/2)^{|S_i ∩ S_j|}, and set μ(F) := log_2((Σ_i λ_i(K))² / Σ_i λ_i(K)²), the log of the stable rank of K(F). Then (i) for every monotone DNF F on N variables with s terms, μ(F) ≤ 4·log_2(2N+s); (ii) for the k-CLIQUE_v indicator on N=C(v,2) edge variables with k=⌈log_2 v⌉, in its canonical minterm DNF (one term per k-clique of K_v, support = edge set of the clique), μ(F_{k-CLIQUE_v}) ≥ v/4; (iii) μ is submodular under conjunction: μ(F∧G) ≤ μ(F)+μ(G) for any monotone DNFs F,G (writing F∧G in its distributed-out form ∨_{i,j} T_i∧T'_j). A single instance violating (i), (ii), or (iii) refutes the conjecture.

Rationale

The Bonami–Beckner kernel at ρ=1/2 admits a per-coordinate rank-2 factorization (f(a,b)=$2^{\{-a-b\}(3/2)}$\{ab\} with det>0), so K(F)=⊗_k K_k restricted to the chosen supports — a hypercontractive Gram structure that controls effective dimensionality of the bottom monomial layer; poly-size DNFs cannot escape the trivial stable-rank ≤ s ≤ poly(N) bound, giving (i). The rigid intersection lattice of k-clique edge sets (two k-cliques sharing j vertices share C(j,2) edges, partitioning the s²=C(v,k)² pairs into k+1 strongly-regular orbits) injects v many independent spectral components into K, driving stable rank to ≥ 2^{v/4}. Submodularity under conjunction follows from tensor structure on disjoint variables (equality) and Cauchy–Schwarz contraction of merged Gram blocks on shared variables; crucially μ is a syntactic invariant of the DNF, dodging NATURAL_PROOFS since computing it requires the formula, not the truth table.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [1806.02734v3] Spectral lower bounds for the orthogonal and projective ranks of a graph - [1801.08709v2] Adaptive Lower Bound for Testing Monotonicity on the Line - [1110.6594v1] Around Operator Monotone Functions - [2501.12063v2] Gram-like matrix preserving extensions and completions of noncommutative polynomials - [2510.18132v1] Beurling Nyman Geometry and Gram Matrix Structure, Ladder Density and Polynomial Decay via Mellin Smoothing

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d4d4e1b2.py", line 166, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d4d4e1b2.py", line 107, in run_trial
    mu = compute_mu(K)
         ^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d4d4e1b2.py", line 56, in compute_mu
    eigenvalues = matrix_eigenvalues(K)
                  ^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d4d4e1b2.py", line 33, in matrix_eigenv
    raise ValueError("Matrix too large for this implementation")
ValueError: Matrix too large for this implementation
```

Judge reasoning

The test crashed with a ValueError (‘Matrix too large for this implementation’) before any instance could be evaluated, so no support_fraction was produced and the pre-registered criterion is neither met nor falsified. | next: Replace the bespoke matrix_eigenvalues routine with numpy.linalg.eigvalsh (or scipy) so the Gram matrices for Families A/B/C can be diagonalized at the required sizes, then rerun the 270+ instance battery.

S-Transform Free-Mult Defect Lower-Bounds DISJ Communication

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Free probability — Voiculescu's S-transform and free multiplicative convolution ⊠ of compactly supported measures on R_+ (Voiculescu 1987; Bercovici–Voiculescu 1992; Haagerup–Schultz 2007). Multiplicative free convolution is distinct from additive (R-transform) free convolution previously attempted on graph Laplacians; an arXiv search for 'S-transform' AND ('communication complexity' OR 'sign rank' OR 'discrepancy method') returns 0 direct hits with <5 adjacent papers, all on Wigner-type ensembles in QIT, never on Boolean communication matrices. × Randomized two-party communication complexity CC_R(M) of DISJOINTNESS and other Boolean matrices M ∈ {0,1}^{N×N} (Razborov 1992 Ω(n) regime), with N = 2^n; the τ(M) tensor-invariant program for COMM_DISJ.
- **Recorded**: 2026-05-17 20:26 UTC
- **Entry ID**: `6c8daf2ab19d`

Statement

For Boolean M ∈ {0,1}^{N×N} let Q_M := (1/N) · M^T M and let σ_M be its empirical spectral measure, normalized to unit total mass and unit first moment (m_1=1) by rescaling. Compute moments m_k = (1/N) tr((Q_M/m_1)^k) for k=1..K with K=⌈log_2 N⌉+2; form the truncated ψ-series ψ(z)=Σ m_k z^k, invert it to χ(z)=ψ^{-1}(z) by Lagrange inversion, set S_M(z)=(1+z)·χ(z)/z, and obtain the moments {μ_k} of the free multiplicative self-convolution σ_M⊠σ_M from S_{⊠}(z)=S_M(z)^2 by re-inverting. Define the free-mult defect δ(M) := |μ_2 − 2·m_2 − 1| / (1+m_2). Conjecture: there is an absolute constant c ∈ [1/64, 1/4] such that (i) for every Boolean M, CC_R(M) ≥ c · log_2(N) · δ(M); (ii) δ(M_DISJ_n) ≥ 1/8 for all n ≥ 3. A single Boolean M with U(M) < c·log_2(N)·δ(M) (where U is any certified CC upper bound), or a single n ≥ 3 with δ(M_DISJ_n) < 1/8, refutes the conjecture.

Rationale

Multiplicative free convolution σ⊠σ encodes asymptotic FREENESS between the row-space and column-space spectral projections of M; deviation from the freeness identity μ_2 = 2m_2 + 1 detects multiplicative coupling between input partitions, which is exactly the structural property Razborov exploits to force long bipartite rectangles in DISJ. Unlike additive R-transform invariants (previously tried on graph Laplacians), the S-transform is intrinsically multiplicative and respects the row × column tensor structure of CC matrices. The bridge is barrier-shielded: Lagrange inversion is non-polynomial in the entries, the invariant is RELATIONAL (a comparison between two measures, not a single truth-table functional), CC lower bounds are not subject to the natural-proofs barrier, and the construction uses no ring-only operations on a Boolean ring extension, so it does not algebrize.

Novelty

- Judge: `NOVEL` over 13 arXiv hits

Top hits consulted: - [2004.07191v1] Cauchy-Stieltjes kernels families and free multiplicative convolution - [1407.6169v3] Multiplicative Complexity of Vector Valued Boolean Functions - [2511.02758v3] Finite free probability and S transforms of Jacobi processes - [2511.07739v3] A Lower Bound for the Fourier Entropy of Boolean Functions on the Biased Hypercube - [1806.02734v3] Spectral lower bounds for the orthogonal and projective ranks of a graph

Empirical Test

- exit code: `0`, elapsed: `94.45s`


```
16, "conjecture_holds": true, "counterexample": "", "seed": 503}
TRIAL: {"metric_name": "free_mult_defect", "metric_value": 1.2174309166311343, "instances_tested": 1
TRIAL: {"metric_name": "free_mult_defect", "metric_value": 1.218306819204241, "instances_tested": 16
TRIAL: {"metric_name": "free_mult_defect", "metric_value": 1.2381510000250828, "instances_tested": 1
1 matrix: U < 0.05 log2(N) delta: n=3, U=0.0, delta=0.9908707865168539", "seed": 631}
TRIAL: {"metric_name": "free_mult_defect", "metric_value": 1.2508519087688272, "instances_tested": 1
TRIAL: {"metric_name": "free_mult_defect", "metric_value": 1.242301935238181, "instances_tested": 16
TRIAL: {"metric_name": "free_mult_defect", "metric_value": 1.2526578087928977, "instances_tested": 1
TRIAL: {"metric_name": "free_mult_defect", "metric_value": 1.219413383084314, "instances_tested": 16
TRIAL: {"metric_name": "free_mult_defect", "metric_value": 1.2409823554557478, "instances_tested": 1
TRIAL: {"metric_name": "free_mult_defect", "metric_value": 1.215808549927829, "instances_tested": 16
RESULT: FALSIFIED counterexample=Rank-1 matrix: U < 0.05 log2(N) delta: n=3, U=0.0, delta=0.75694444
```


Judge reasoning

Safety rail: critic_challenge_falsified | original: The test's RESULT line reports FALSIFIED with explicit counterexamples (Identity and Rank-1 matrices at n=3,4,5 having U=0 but δ≈0.75–0.99), violating | next: Restrict the conjecture to Boolean matrices of high combinatorial rank (e.g., exclude low-rank/structured M) or replace U with a lower bound on CC_R rather than any certified upper bound.

Fekete Capacity of Laplacian Spectrum Upper-Bounds Max-Cut SoS-2 Gap

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Logarithmic potential theory — Fekete transfinite diameter / Vandermonde capacity of finite point sets on R, in the Saff–Totik / Ransford / Hardin–Saff tradition (equilibrium measures, Robin constants, weighted log-energy). This corner of classical analysis has essentially no presence in SoS / CSP integrality-gap literature (arXiv 'transfinite diameter' AND ('sum of squares' OR 'Lasserre' OR 'Max-Cut') returns 0 direct hits, with <5 adjacent random-matrix papers on Wigner-type spectral statistics). × Degree-2 Sum-of-Squares / Lasserre integrality gap for Max-Cut on small graphs, accessed through the Delorme–Poljak eigenvalue upper bound SDP_2(G) ≤ (n/4)·λ_n(L_G) (which coincides with degree-2 SoS for vertex-transitive graphs and is a valid SoS-2 upper bound in general). This is the Goemans–Williamson / Schoenebeck–Khot–Vishnoi / Barak–Steurer regime targeted by SOS_HIERARCHY.
- **Recorded**: 2026-05-17 20:47 UTC
- **Entry ID**: `999ba4b45fab`

Statement

For a connected graph G on n vertices with Laplacian L_G having nontrivial eigenvalues $0 < \lambda_2 \le \ldots \le \lambda_n$, let $\tilde{\sigma}(G) = \{\lambda_2/\lambda_n, \ldots, \lambda_{n-1}/\lambda_n, 1\}$ be the spectrum rescaled into (0,1], and define the normalized Fekete capacity $\mathrm{Cap}(G) := (\prod_{2 \le i<j \le n} |\tilde{\lambda}_i - \tilde{\lambda}_j|)^{2/((n-1)(n-2))} \in [0,1]$ (the transfinite diameter of the rescaled spectrum, taken as 0 when any pair coincides exactly). Let $\mathrm{gap}(G) := n\cdot\lambda_n / (4\cdot\mathrm{MaxCut}(G)) - 1 \ge 0$ be the Delorme–Poljak / SoS-2 multiplicative integrality gap. Conjecture: there is an absolute constant $C \le 1$ such that for every connected graph G on $6 \le n \le 16$ vertices, $\mathrm{gap}(G) \le C \cdot \mathrm{Cap}(G)$; equivalently, any nonzero SoS-2 integrality gap forces the rescaled Laplacian spectrum to have positive transfinite diameter, with the gap controlled linearly by it. A single graph G with $\mathrm{gap}(G) > C \cdot \mathrm{Cap}(G)$ by additive margin > 0.01 refutes the conjecture.

Rationale

Bipartite-like graphs (K_{n/2,n/2}, balanced complete multipartite) saturate the Delorme–Poljak bound (gap = 0) AND have multiply-repeated Laplacian eigenvalues (Cap = 0), while expander 3-regular graphs have a well-spread spectrum (Cap > 0) AND a positive gap (≈ 0.087); the conjecture quantifies this co-occurrence by bounding the SoS-2 gap above by the geometric-mean spectral-gap product, a classical potential-theoretic invariant from Saff–Totik that has never been used as a hardness functional. The invariant is computed on the *graph* (a structural object) — not on a truth table — so it is shielded from NATURAL_PROOFS; it uses logarithms (non-ring), so it is shielded from ALGEBRIZATION; and it concerns concrete graphs, not oracles, so it is shielded from RELATIVIZATION. The bridge sharpens existing Delorme–Poljak / Goemans–Williamson analysis by replacing the single λ_n with the full Vandermonde-style multiplicative structure of σ(L_G).

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [1107.1630v3] On the Integrality Gap of the Subtour LP for the 1,2-TSP - [1207.5696v1] Beyond Max-Cut: λ-Extendible Properties Parameterized Above the Poljak-Turzík Bound - [2205.10968v4] Eigenvalue bounds of the Kirchhoff Laplacian - [1210.8059v3] Equidistribution estimates for Fekete points on complex manifolds - [1301.6896v1] Laplacians on periodic discrete graphs

Empirical Test

- exit code: `0`, elapsed: `0.15s`

```
6, "conjecture_holds": false, "counterexample": "n=8, gap=0.228280467467467, cap=0.1775420094450226,
TRIAL: {"metric_name": "gap_minus_cap", "metric_value": 0.0, "metric_std": 0.0, "instances_tested":
TRIAL: {"metric_name": "gap_minus_cap", "metric_value": 0.0, "metric_std": 0.0, "instances_tested":
TRIAL: {"metric_name": "gap_minus_cap", "metric_value": 0.0, "metric_std": 0.0, "instances_tested":
TRIAL: {"metric_name": "gap_minus_cap", "metric_value": 0.0, "metric_std": 0.0, "instances_tested":
TRIAL: {"metric_name": "gap_minus_cap", "metric_value": 0.0, "metric_std": 0.0, "instances_tested":
```

Judge reasoning

Safety rail: critic_challenge_falsified | original: Multiple instances violate gap(G) ≤ Cap(G) + 0.01 by large margins; e.g., n=8 with gap≈0.2587, Cap≈0.1688 (margin ≈0.09), and degenerate cases like se | next: Replace Fekete capacity with a degeneracy-robust separation surrogate (e.g., min pairwise gap regularized as log(1/(ε+min|λ̃_i−λ̃_j|))) or restrict to graphs with simple Laplacian spectra before re-testing the gap bound.

Polymer Cluster-Expansion Convergence Radius Bounds Tree-Res Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Statistical mechanics — Fernandez–Procacci / Kotecký–Preiss cluster-expansion convergence radius for abstract polymer models (Mayer 1937; Kotecký–Preiss 1986; Fernandez–Procacci 2007; Bissacot–Fernandez–Procacci 2010). An analytic structural invariant of finite graphs essentially absent from proof complexity outside loose Lovász-Local-Lemma analogies; an arXiv search 'cluster expansion' AND ('Resolution refutation' OR 'DPLL' OR 'Frege' OR 'proof complexity') returns no direct hits and <5 adjacent papers, none using FP fixed-point activities as a CNF invariant. × Tree-like Resolution / DPLL refutation tree size t*(F) for unsatisfiable 3-CNFs, in the Ben-Sasson–Wigderson–Krajíček bounded-arithmetic regime where ¬F is Σ^b_0 and log_2 t*(F) controls V^0-witnessing complexity — a Frege/Extended-Frege stepping stone via the Pudlák–Buss feasible-interpolation and Krajíček EF-depth simulation chain.
- **Recorded**: 2026-05-17 21:20 UTC

- **Entry ID**: `3905f361f983`

Statement

For an unsatisfiable 3-CNF F on n variables with clauses $C_1,\ldots,C_m$, build the clause-overlap graph G(F) (vertices = clauses; edge iff the two clauses share a variable) and let $\mu(F) \in (0,1]^m$ be the unique fixed point of the Fernandez–Procacci iteration $\mu_v \leftarrow 1/(1 + \Sigma_{u \sim v} \mu_u)$ initialised at $\mu_v^{(0)}=1$; define the polymer convergence defect $\Lambda(F) := -\Sigma_v \log_2 \mu_v$. We conjecture there is an absolute constant $c \geq 1/8$ such that $\log_2 t(F) \geq c \cdot \Lambda(F)$ for every unsatisfiable 3-CNF F, with the bound near-tight on random 3-CNFs at $\alpha \approx 4.26$. A single unsatisfiable F with $\log_2 t(F) < (1/8)\cdot\Lambda(F)$ falsifies the conjecture.

Rationale

Cluster-expansion convergence is the analytic dual of clause-overlap expansion: the FP activities μ_v contract precisely when local clause interactions admit a convergent Mayer series, mirroring the same expansion mechanism underlying Ben-Sasson–Wigderson width lower bounds — but packaged as a single-number invariant of the SYNTACTIC formula, not of any truth-table, so it dodges the Razborov–Rudich naturalization route. The construction is non-arithmetizing (the fixed point is a graph-combinatorial invariant of the CNF symbol level, not a polynomial extension of the Boolean ring), avoiding the algebrization wall hit by GCT cubic-form attempts. If correct, Λ(F) would supply a physics-style 'free-energy' depth measure for DPLL and, via interpolation, the first quantitative structural bridge from cavity-method analysis to Frege/EF lower bounds.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [1705.04385v2] A correction to a remark in a paper by Procacci and Yuhjtman: new lower bounds for the convergence radius of the virial - [0605041v2] Cluster expansion for abstract polymer models. New bounds from an old approach - [1002.3261v1] On the convergence of cluster expansions for polymer gases - [1904.03250v1] Velocity jump in the crack propagation induced on a semi-crystalline polymer sheet by constant-speed stretching - [0906.0693v3] An improved lower bound on the counterfeit coins problem

Empirical Test

- exit code: `0`, elapsed: `0.35s`

```
es_tested': 0, 'conjecture_holds': False, 'counterexample': 'No valid instances generated'}
TRIAL: {'metric_name': 'r(F)', 'metric_value': 0.0, 'instances_tested': 0, 'conjecture_holds': False
TRIAL: {'metric_name': 'r(F)', 'metric_value': 0.0, 'instances_tested': 0, 'conjecture_holds': False
TRIAL: {'metric_name': 'r(F)', 'metric_value': 0.0, 'instances_tested': 0, 'conjecture_holds': False
TRIAL: {'metric_name': 'r(F)', 'metric_value': 0.0, 'instances_tested': 0, 'conjecture_holds': False
TRIAL: {'metric_name': 'r(F)', 'metric_value': 0.0, 'instances_tested': 0, 'conjecture_holds': False
TRIAL: {'metric_name': 'r(F)', 'metric_value': 0.0, 'instances_tested': 0, 'conjecture_holds': False
TRIAL: {'metric_name': 'r(F)', 'metric_value': 0.0, 'instances_tested': 0, 'conjecture_holds': False
TRIAL: {'metric_name': 'r(F)', 'metric_value': 0.0, 'instances_tested': 0, 'conjecture_holds': False
TRIAL: {'metric_name': 'r(F)', 'metric_value': 0.0, 'instances_tested': 0, 'conjecture_holds': False
TRIAL: {'metric_name': 'r(F)', 'metric_value': 0.0, 'instances_tested': 0, 'conjecture_holds': False
TRIAL: {'metric_name': 'r(F)', 'metric_value': 0.0, 'instances_tested': 0, 'conjecture_holds': False
TRIAL: {'metric_name': 'r(F)', 'metric_value': 0.0, 'instances_tested': 0, 'conjecture_holds': False
RESULT: FALSIFIED counterexample=No valid instances generated first_failing_seed=11
```

Judge reasoning

The test generated zero valid instances across all 210 trials (instances_tested: 0, metric_value: 0.0), so the conjecture was never empirically evaluated. The 'FALSIFIED' RESULT line reflects a generator/construction failure, not a genuine counterexample to the bound. | next: Fix the unsatisfiable 3-CNF generator (e.g., sample random 3-CNFs at $\alpha=4.26$ and filter via a SAT solver, or use known unsat constructions like PHP_n+1^n encodings) so that t*(F) and Λ(F) can actually be computed for all 210

Fatgraph Genus of Canonical Clause Ribbon Lower-Bounds Resolution Width

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Topological graph theory of ribbon graphs / fatgraphs (Penner 1987; Mulase-Penkava 1998; Bollobás-Riordan 2002; Loebl-Moffatt 2010) — the oriented surface embedding genus of a graph with prescribed cyclic edge rotation at every vertex, computed combinatorially in O(|E|) by tracing boundary curves and applying Euler's formula. This corner of combinatorial topology has essentially zero presence in proof complexity (arXiv 'ribbon graph' AND ('resolution refutation' OR 'proof complexity' OR 'Frege') returns 0 direct hits; <5 adjacent papers, none using fatgraph genus as a CNF invariant). × Resolution refutation width w_Res(F) of unsatisfiable 3-CNFs, viewed in the Ben-Sasson–Wigderson regime where w_Res lower-bounds tree-Resolution size and serves as the canonical Frege-depth stepping stone in the Cook–Reckhow–Krajíček bounded-arithmetic framework (¬F is Σ^b_0, $\log_2$ of tree-Resolution size controls V^0-witnessing complexity).
- **Recorded**: 2026-05-17 21:53 UTC
- **Entry ID**: `523a031cbc3e`

Statement

Given an unsatisfiable 3-CNF F on n variables and m clauses, build the canonical ribbon graph R(F) as follows: introduce a vertex u_C for each clause C and a vertex w_v for each variable v; place a half-edge (u_C, w_v) for every clause-variable incidence; at u_C use the cyclic rotation σ_C = (sign(l_1)·idx(l_1), sign(l_2)·idx(l_2), sign(l_3)·idx(l_3)) sorted ascending by signed index, and at w_v use the cyclic rotation that alternates positive and negative occurrences ordered by clause index. Trace boundary cycles by alternating σ-then-τ application on half-edges; let F(R) be the number of traced cycles and define $g(F) := (2 - (m+n) + 3m - F(R))/2$ (orientable genus of the canonical embedding). Conjecture: there is an absolute constant $c > 0$ such that for every unsatisfiable 3-CNF F, the Resolution refutation width satisfies $w\_Res(F) \geq c \cdot \log_2(1 + g(F))$; a single unsat 3-CNF F with $w\_Res(F) < \lfloor 0.25 \cdot \log_2(1 + g(F)) \rfloor$ falsifies the conjecture.

Rationale

Ribbon graph genus encodes the obstruction to drawing the clause-variable incidence as a planar (sphere-embedded) structure under a syntactically prescribed local rotation system; resolution refutations geometrically correspond to face-collapse moves on this surface, so genus is a combinatorial certificate of the minimum boundary-tracing complexity any refutation must witness. Unlike the Hochster-regularity / Hodge-Cheeger attempts in the blacklist, this invariant is purely combinatorial (face tracing) and avoids commutative-algebra blow-up, while remaining shielded from natural-proofs barriers because it is a property of the CNF SYNTAX (clause rotations including signs) not of the truth table; the same Boolean function admits many CNF representations with different g(F).

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [1804.02478v5] Complexity of the CNF-satisfiability problem - [1905.05290v2] Graph Width Measures for CNF-Encodings with Auxiliary Variables - [1905.10867v2] Regular resolution for CNFs with almost bounded one-sided treewidth - [1905.12372v1] Resolution Lower Bounds for Refutation Statements - [2511.07739v3] A Lower Bound for the Fourier Entropy of Boolean Functions on the Biased Hypercube

Empirical Test

- exit code: 1, elapsed: 1.47s

```
: ''}
TRIAL: {'metric_name': 'resolution_width_lower_bound', 'metric_value': 0.0, 'instances_tested': 1, '
CNF generated with seed 53'}
TRIAL: {'metric_name': 'resolution_width_lower_bound', 'metric_value': 7.076815597050831, 'instance
TRIAL: {'metric_name': 'resolution_width_lower_bound', 'metric_value': 7.22881869049588, 'instances_
TRIAL: {'metric_name': 'resolution_width_lower_bound', 'metric_value': 7.022367813028454, 'instance
TRIAL: {'metric_name': 'resolution_width_lower_bound', 'metric_value': 0.0, 'instances_tested': 1, '
CNF generated with seed 127'}
TRIAL: {'metric_name': 'resolution_width_lower_bound', 'metric_value': 0.0, 'instances_tested': 1, '
CNF generated with seed 149'}
TRIAL: {'metric_name': 'resolution_width_lower_bound', 'metric_value': 7.022367813028454, 'instance
TRIAL: {'metric_name': 'resolution_width_lower_bound', 'metric_value': 6.906890595608519, 'instance

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9da09eeb.py", line 147, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9da09eeb.py", line 127, in run_trial
    w_res = compute_resolution_width(clauses, n)
            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9da09eeb.py", line 99, in compute_resol
    width = dp[frozenset(new_subset)] + 1
               ^^^^^^^^^^^^^^^^^^^^^
TypeError: unhashable type: 'list'
```

Judge reasoning

The test crashed with a TypeError (unhashable list in the width-DP) before completing the pre-registered 30 seeds × 3 densities × 6 sizes sweep plus Tseitin instances, so the support condition cannot be evaluated. Partial trials shown are dominated by satisfiable-instance skips and a handful of unsat passes, which is insufficient evidence either way. | next: Fix the width-DP bug (convert subset to frozenset/tuple before hashing) and rerun the full pre-registered sweep including Tseitin instances

---

Forman-Ricci Min-Curvature of Term-Overlap Graph Lower-Bounds Monotone k-CLIQUE DNF

- **Verdict**: `FALSIFIED`
- **Bridge**: Forman's combinatorial Ricci curvature on weighted 1-skeleta (Forman 2003, 'Bochner's method for cell complexes and combinatorial Ricci curvature'; Sreejith-Mukherjee-Sandhu-Saucan-Jost 2016 on network curvature). Distinct from Ollivier-Ricci (Skenderi 2025): Forman uses parallel-face / Bochner-Weitzenböck local sums of vertex and edge weights, not Wasserstein optimal transport. An arXiv search for 'Forman Ricci' AND ('monotone circuit' OR 'DNF' OR 'proof complexity' OR 'Razborov') returns 0 direct hits and <5 adjacent papers, all on biological/social networks — never on Boolean-function DNFs. × Monotone DNF leaf size / formula size for the k-CLIQUE indicator on K_v (n = v(v-1)/2 edge variables) in the Razborov 1985 / Alon-Boppana / Andreev regime; the Cook-Reckhow-Krajicek bounded-arithmetic

stepping stone where a submodular DNF measure that is small on poly-size DNFs but $\Omega(v)$ on k-CLIQUE would witness a V^0-style separation by Buss-Pudlak-style approximator counting.

- **Recorded**: 2026-05-17 22:25 UTC
- **Entry ID**: `b0a4fb5d3039`

Statement

For a monotone DNF $F = \vee_{i=1}^{s} T_i$ (terms $T_i \subseteq [N]$) with at least one term of size ≥ 1, build the weighted term-overlap graph G_F on vertex set $\{1,\dots,s\}$: place an edge ij iff $|T_i \cap T_j| \geq 1$, assign vertex weight $w_i = |T_i|$ and edge weight $w_{ij} = |T_i \cap T_j|$, and define the Forman-Ricci curvature $\mathrm{Ric}_F(ij) = w_{ij}\cdot[\, w_i/w_{ij} + w_j/w_{ij} - \Sigma_{k\sim i,\, k\neq j}\, w_i/\sqrt{w_{ij}\cdot w_{ik}} - \Sigma_{k\sim j,\, k\neq i}\, w_j/\sqrt{w_{ij}\cdot w_{jk}}\,]$. Let $\mu(F) := \log_2(1 + \max\{0, -\min_e \mathrm{Ric}_F(e)\})$ (with $\mu = 0$ if G_F has no edges). We conjecture: (i) for any pair of monotone DNFs F,G, $\mu(F \wedge G) \leq \mu(F) + \mu(G) + \log_2(1 + N)$ (approximate submodularity under conjunction); (ii) for every monotone DNF F with $|F| \leq N^c$ terms, $\mu(F) \leq 6c\cdot\log_2(1 + N)$; and (iii) the canonical minterm DNF F^*_v of the k-CLIQUE indicator on K_v with $k = \lceil \log_2 v \rceil$ satisfies $\mu(F^*_v) \geq v/4$. A single instance violating any of (i),(ii),(iii) refutes the conjecture.

Rationale

The Razborov approximator method replaces ∧ and ∨ with closed operations on 'small CC^k objects' and bounds the error count; the right measure must be submodular under ∧ and saturate on clique. Forman-Ricci collapses each term-overlap edge into a single weighted Bochner-Weitzenböck term, so global negativity of the minimum curvature is forced precisely when the term graph contains a high-degree dense neighborhood around a 'sunflower core' — exactly the structural obstruction that makes clique indicators hard to approximate. Because μ is computed on the DNF graph (not the truth table) and uses √-weighted sums (not ring polynomial extensions), it is shielded from both NATURAL_PROOFS and ALGEBRIZATION.

Novelty

- Judge: `NOVEL` over 6 arXiv hits

Top hits consulted: - [2103.10093v1] Conjectures on Convergence and Scalar Curvature - [2403.15973v1] Isoperimetric profile function comparisons with Integral Ricci curvature bounds - [1809.07698v1] Discrete Ricci curvatures for directed networks - [2304.02672v2] The Log-Sobolev inequality for a submanifold in manifolds with asymptotic non-negative intermediate Ricci curvature - [1909.00234v1] The spectrum of a class of uniform hypergraphs

Empirical Test

- exit code: `0`, elapsed: `0.03s`

```
ances_tested': 1, 'conjecture_holds': False, 'counterexample': 'Regime A: v=4, mu=0 < v/4=1.0', 'seed
TRIAL: {'metric_name': 'mu', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_holds': False, '
TRIAL: {'metric_name': 'mu', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_holds': False, '
TRIAL: {'metric_name': 'mu', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_holds': False, '
TRIAL: {'metric_name': 'mu', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_holds': False, '
TRIAL: {'metric_name': 'mu', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_holds': False, '
TRIAL: {'metric_name': 'mu', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_holds': False, '
TRIAL: {'metric_name': 'mu', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_holds': False, '
TRIAL: {'metric_name': 'mu', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_holds': False, '
TRIAL: {'metric_name': 'mu', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_holds': False, '
TRIAL: {'metric_name': 'mu', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_holds': False, '
TRIAL: {'metric_name': 'mu', 'metric_value': 0.0, 'instances_tested': 1, 'conjecture_holds': False, '
RESULT: FALSIFIED counterexample="Regime A: v=4, mu=0 < v/4=1.0" first_failing_seed=11
```

Judge reasoning

Regime (A) consistently yields $\mu(F^*_v) = 0$ for v=4, violating clause (iii)'s requirement that $\mu(F^*_v) \geq v/4 = 1.0$. The test's RESULT line reports FALSIFIED with a reproducible counterexample at first_failing_seed=11. | next: Replace the log_2(1+max(0,-min Ric)) aggregator with a curvature-gap functional that scales with clique number (e.g., based on spectral gap of the term-overlap Laplacian) before reattempting a k-CLIQUE lower bound.

Bourgain-Tzafriri Sub-Column Spectral Excess Bounds DISJ CC_R

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Bourgain–Tzafriri / Kashin restricted invertibility theory — average operator norm of random k-column submatrices of a centered sign matrix, in the Bourgain–Tzafriri 1987 / Kashin 1977 / Spielman–Srivastava 2010 tradition of restricted invertibility and column-subset selection. An arXiv search for 'Bourgain Tzafriri' AND ('communication complexity' OR 'DISJOINTNESS') returns 0 direct hits and <5 adjacent papers, all on Forster-style spectral sign-rank rather than sub-column dispersion. × Randomized two-party communication complexity CC_R(M) of N×N Boolean matrices, with target M_DISJ_n on N=2^n (Razborov 1992 Ω(n) regime), as an instance of unsaturated communication bounds via dispersion methods.
- **Recorded**: 2026-05-17 22:43 UTC
- **Entry ID**: `14df14ed8261`

Statement

Center $M \in \{0,1\}^{N\times N}$ to $\tilde{M} = 2M - J \in \{\pm 1\}^{N\times N}$. Fix $k = \lceil \log_2 N \rceil$ and let $S_1,\ldots,S_{30} \subset [N]$ be uniformly random column subsets of size k drawn with seeds 1..30. Define the sub-column spectral excess $\xi(M) := (1/30)\cdot\Sigma_{s=1}^{30}(\|\tilde{M}|_{S_s}\|_{op}^2 / N - 1)$, where $\tilde{M}|_S$ is the N×k column submatrix. We conjecture that for every Boolean matrix M with $N \leq 32$: (i) $\xi(M) \geq 0$; (ii) $CC_R(M) \geq \lfloor \log_2(1 + N\cdot\xi(M)/k)\rfloor - 1$; (iii) $\xi(M_{PARITY\_n}) \leq 0.05$ (so the bound gives O(1), matching $CC_R = O(1)$) and $\xi(M_{DISJ\_n}) \geq 0.5\cdot k/n$ at n=3,4,5 (so the bound delivers Ω(log N) for DISJ). A single seed-30 ensemble at any $n \in \{3,4,5\}$ with $\xi(M_{DISJ\_n}) < 0.3\cdot k/n$ or $\xi(M_{PARITY\_n}) > 0.1$ falsifies the conjecture.

Rationale

Sub-column spectral excess measures the average deviation of k-column Gram spectra from the orthogonal (Hadamard) isotropic baseline, where the baseline $\|\tilde{M}|_S\|_{op}^2/N = 1$ holds exactly for parity-XOR matrices because their columns are orthogonal. Structured matrices like M_DISJ_n have correlated columns whose Gram matrix has a dominant 'all-disjoint-tail' eigenvector, forcing $\|\tilde{M}|_S\|_{op}^2 > N$ on most subsets. Crucially the invariant is RELATIONAL (an empirical spectral statistic over random column subsamples, not a single-number truth-table functional) and is not large in the natural-proofs sense (random Boolean matrices have $\xi \approx k/N \to 0$), so it sidesteps both the Razborov–Rudich barrier and the algebrization filter, while computationally reducing to 30 small SVDs.

Novelty

- Judge: `NOVEL` over 11 arXiv hits

Top hits consulted: - [2208.05223v1] Modular Bourgain-Tzafriri Restricted Invertibility Conjectures and Johnson-Lindenstrauss Flattening Conjecture - [2503.20972v3] p-adic Grothendieck Inequality, p-adic Johnson-Lindenstrauss Flattening and p-adic Bourgain-Tzafriri Restricted Invertib - [0111062v2] One-way communication complexity and the Neciporuk lower bound on formula size - [1005.0979v1] Supersymmetry in Random Matrix Theory - [0904.1432v2] Applications of random matrix theory to condensed matter and optical physics

Empirical Test

- exit code: `0`, elapsed: `0.33s`


```
j", "metric_value": 119.11555989583333, "instances_tested": 270, "conjecture_holds": false, "counter
TRIAL: {"seed": 503, "metric_name": "xi_disj", "metric_value": 114.68014322916667, "instances_tested
TRIAL: {"seed": 547, "metric_name": "xi_disj", "metric_value": 108.26382378472222, "instances_tested
TRIAL: {"seed": 593, "metric_name": "xi_disj", "metric_value": 161.8962890625, "instances_tested": 2
TRIAL: {"seed": 631, "metric_name": "xi_disj", "metric_value": 109.49559461805556, "instances_tested
TRIAL: {"seed": 677, "metric_name": "xi_disj", "metric_value": 95.28934461805557, "instances_tested"
TRIAL: {"seed": 727, "metric_name": "xi_disj", "metric_value": 132.66538628472222, "instances_tested
TRIAL: {"seed": 773, "metric_name": "xi_disj", "metric_value": 137.6169487847222, "instances_tested"
TRIAL: {"seed": 821, "metric_name": "xi_disj", "metric_value": 112.67233072916667, "instances_tested
TRIAL: {"seed": 877, "metric_name": "xi_disj", "metric_value": 133.46503906249998, "instances_tested
TRIAL: {"seed": 929, "metric_name": "xi_disj", "metric_value": 139.62198350694447, "instances_tested
RESULT: FALSIFIED counterexample="xi_parity=25.05 > 0.05" first_failing_seed=11
```


Judge reasoning

Safety rail: critic_challenge_falsified | original: The test's RESULT line reports FALSIFIED with counterexample xi_parity=25.05 > 0.05 on seed 11, and all 30 seeds show xi_parity in the 14–25 range, va | next: Replace raw spectral excess with a PARITY-insensitive functional (e.g., subtract the expected operator norm of a random ±1 N×k matrix, ≈ $(\sqrt{N}+\sqrt{k})^2$, or use the L1→L2 sub-column norm) so that ξ(PARITY) ≈ 0 while ξ(DISJ) remains Ω(k/n).

Trivial-Isotype Mass Gap Separates Perm from Padded Det

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Schur-Weyl / Specht-isotype decomposition of the regular representation of S_n applied to the permutation-diagonal coefficient vector of homogeneous polynomials — an algebraic-combinatorial structural invariant in the Stembridge-Greene-James-Kerber tradition; an arXiv search 'Specht isotype' AND ('determinantal complexity' OR 'orbit closure' OR 'padded determinant') returns 0 direct hits, with <5 adjacent papers (none mapping Specht-isotype mass to GCT obstructions of perm_n). × Determinantal complexity dc(perm_n) in the Mulmuley-Sohoni GCT_DET_PERM regime — the smallest m such that perm_n lies in the GL_{n²}-orbit closure of □(y)^{n-m} · det_m(L(y)) where L is an m×m matrix of linear forms in y={y_{ij}}_{1≤i,j≤n}, the canonical Landsberg 2017 / Mignon-Ressayre target for super-polynomial det-vs-perm separation.
- **Recorded**: 2026-05-17 23:19 UTC
- **Entry ID**: `2d98562b9959`

Statement

For homogeneous f ∈ R[y_{ij}]*{1≤i,j≤n} of degree n, let ν(f) ∈ R^{S_n} be the permutation-diagonal vector ν_σ(f) = [Π_i y_{i,σ(i)}] f, and set ρ(f) = (Σ_σ ν_σ)² / (n! · Σ_σ ν_σ²) when ν(f) ≠ 0 (the squared cosine of ν(f) with the trivial-isotype direction of the regular S_n-representation). Then (i) ρ(perm_n) = 1 (saturation), and (ii) for every m×m matrix L of linear forms L*{ab}(y) and every linear form □(y) in the n² variables, with 2 ≤ m ≤ n−1, the padded form f_{L,□} = □(y)^{n-m} · det_m(L(y)) satisfies ρ(f_{L,□}) ≤ 1 − 1/n. A single pair (L, □) achieving ρ(f_{L,□}) > 1 − 1/n refutes the conjecture.

Rationale

Perm_n uniquely saturates the trivial-isotype of R^{S_n} because its permutation-diagonal coefficient vector is the all-ones direction; det_n analogously saturates the sign-isotype. Padded det_m factors through an m×m antisymmetric tensor (sign-isotype of S_m ↪ S_n) tensored with a symmetric □^{n-m} factor, and by a Pieri/branching argument the induced direction in R^{S_n} cannot reach the trivial saturation when m < n. Because ρ is computed from permutation-diagonal coefficients via finite Gaussian-elimination steps (no truth-table evaluation), the invariant is RELATIONAL on the polynomial structure and shielded from Razborov-Rudich (it is not a poly-time-computable property of a truth table) and from algebrization (the underlying claim is a representation-theoretic projection-saturation, not a polynomial identity over a ring extension).

Novelty

- Judge: `NOVEL` over 8 arXiv hits

Top hits consulted: - [1306.1810v6] Matrix orbit closures - [2202.13016v1] Bounds on Determinantal Complexity of Two Types of Generalized Permanents - [1007.3804v4] Symmetric Determinantal Representation of Formulas and Weakly Skew Circuits - [1503.00822v2] Product Ranks of the 3×3 Determinant and Permanent - [1101.2456v3] Polynomial representations and categorifications of Fock Space

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6ba97aec.py", line 144, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6ba97aec.py", line 107, in run_trial
    m_values = [[2], [n//2], [n-1]]
                      ^
UnboundLocalError: cannot access local variable 'n' where it is not associated with a value
```

Judge reasoning

The test crashed with an UnboundLocalError before producing any ρ values, so neither the support nor the falsification condition could be evaluated. | next: Fix the scoping bug (define n before constructing m_values, e.g., inside the n-loop) and rerun the 90-seed sweep.

Eulerian Descent Entropy Separates Perm from Padded Det_m

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Eulerian descent combinatorics on the symmetric group (Worpitzky identity; Stanley EC1 §1.4; Petersen 2015 'Eulerian Numbers') used as a coefficient-binning functional on the permutation-monomial expansion of homogeneous polynomials — distinct from Mahonian/major-index statistics (blacklisted) and from Specht-isotype mass (blacklisted); arXiv 'Eulerian numbers' AND ('determinantal complexity' OR 'GCT' OR 'padded permanent') returns 0 direct hits and <5 adjacent papers, none using descent-level binning as a GCT obstruction. × GCT det-vs-perm padding complexity: determinantal complexity dc(perm_n) — smallest m such that perm_n lies in the GL_{n²}-orbit closure of □(y)^{n−m}·det_m(L(y)) (Mulmuley–Sohoni / Mignon–Ressayre / Landsberg regime), accessed through the permutation-coefficient vector c_σ(f) of polynomial-padded determinants.
- **Recorded**: 2026-05-17 23:46 UTC
- **Entry ID**: `1aaaed379c87`

Statement

For a homogeneous degree-n polynomial f ∈ R[x_{i,j}: 1≤i,j≤n], let c_σ(f) be the coefficient of ∏*i x*{i,σ(i)} in f, define the Eulerian-descent profile E_k(f) := Σ_{σ ∈ S_n, des(σ)=k} c_σ(f)² for k=0,…,n−1, and let ρ(f) := H_2(Ẽ(f)) be the Shannon entropy of Ẽ := E/Σ_k E_k. Conjecture: (i) ρ(perm_n) ≥ 1 for all n ≥ 4; (ii) for every m ≤ ⌊√n⌋, every linear substitution L: R^{m²} → R^{n²}, and every product padding p = ∏*{j=1}^{n−m} ℓ_j of linear forms ℓ_j in the x*{i,j}, ρ(det_m(L(y))·p) ≤ log_2(2m+1). A single instance violating (i) or (ii) refutes the conjecture.

Rationale

Permanents distribute squared coefficient mass uniformly over all of S_n, producing the full Eulerian distribution A(n,k) whose entropy grows like ½ log n; padded det_m of rank-m linear images can populate only m-minor-controlled descent levels (at most ~2m+1 distinct k values per padded orbit), giving an entropy ceiling of log(2m+1). Descent-number binning is a RELATIONAL S_n functional (not a single-number truth-table invariant), shielding it from Razborov–Rudich naturalisation that swallowed prior Stern–Brocot / immanant / Specht-isotype single-number attempts; the construction never extends to a polynomial ring over Boolean variables, sidestepping Aaronson–Wigderson algebrization.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [1503.00822v2] Product Ranks of the 3×3 Determinant and Permanent - [1802.01767v3] Pseudomonads and Descent, PhD Thesis (Chapter 1) - [1410.8628v1] The Colored Eulerian Descent Algebra - [1511.08113v2] Permanent versus determinant, obstructions, and Kronecker coefficients - [1301.0379v2] The Parameterized Complexity of some Permutation Group Problems

Empirical Test

- exit code: `0`, elapsed: `0.03s`

```
jecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'rho(perm_n)', 'metric_value': 1.6418805457246157, 'instances_tested': 5, 'co
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TRIAL: {'metric_name': 'rho(perm_n)', 'metric_value': 1.6418805457246157, 'instances_tested': 5, 'co
TRIAL: {'metric_name': 'rho(perm_n)', 'metric_value': 1.6418805457246157, 'instances_tested': 5, 'co
TRIAL: {'metric_name': 'rho(perm_n)', 'metric_value': 1.6418805457246157, 'instances_tested': 5, 'co
TRIAL: {'metric_name': 'rho(perm_n)', 'metric_value': 1.6418805457246157, 'instances_tested': 5, 'co
TRIAL: {'metric_name': 'rho(perm_n)', 'metric_value': 1.6418805457246157, 'instances_tested': 5, 'co
RESULT: SUPPORTED mean=1.6418805457246157 std=0.0 support_fraction=1.0
```

Judge reasoning

The stdout only reports ρ(perm_n) trials (a deterministic Eulerian-number computation repeated identically), with no evidence that the 450 padded-det trials in part (ii) were executed; the pre-registered support condition requires both halves to pass. Per the critic challenge and the missing adversarial test of ρ(det_m(L(y))·p) ≤ log_2(2m+1), SUPPORTED must be downgraded. | next: Implement and run the part-(ii) sweep: for each n ∈ {4..8}, m ∈ {1,2,⌊√n⌋}, and 30 random seeds sampling linear subst